Genus Budgets and Native Boundary Interfaces for Unknot Hierarchies

Rigorous reductions, source-bound certificates, and a quantitative research program toward quasipolynomial recognition

Research continuation for the ProveIt project
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Abstract

We continue the ProveIt unknot-recognition program at a pinned repository state and examine four pending research packages. The central problem is to turn efficient operations on compressed normal surfaces into a complete hierarchy algorithm with a proved quasipolynomial bound. This article establishes and implements two missing local bridges and proves a structural reduction that explains their limits.

First, cutting unused one-handle cocores in an irreducible manifold gives an exact Euler-deficit budget. For connected boundary of genus g, at most g nonball pieces survive, and ball absorption reduces their attachment network to at most $g-1$ handles. In a torus source this yields either a solid-torus certificate or a unique inherited core. A local tetrahedral argument proves initial empty-pattern admissibility of that core. We implement authenticated diagram wrappers and independent geometric replay, and prove a certificate-size improvement from quadratic to $O(t \log t)$ bits on the specified Fibonacci layered family.

Second, canonical ownership of native boundary edges permits one weighted orbit census to recover complete first-return data, including exact gap lengths and local fibre transport. A compact arithmetic encoding replaces large endpoint masks; a conservation argument verifies pair occupancy without a separate count field. The implementation reconstructs its source independently and is compared against both the maintained compiler and an explicit small-input oracle.

We also derive an output-sensitive unmarked reconstruction bound and give a conditional global complexity theorem that identifies the remaining geometric and quantitative obligations. The package includes proofs, source code, independent checkers, adversarial examples, raw measurements, and reproducible artifacts. These results do not establish general quasipolynomial unknot recognition. Measured regressions, unimplemented geometric outputs, and the distinction between source authentication and numerical summaries are stated explicitly.

Audited baseline. https://github.com/VladimirReshetnikov/ProveIt
Commit `704600626606a9097cfd89d4e3d2bbc63c8dd1cb`.
The accompanying archive contains the complete article sources, executable research additions, a pinned Python source snapshot, and the evidence needed to reproduce the reported comparisons. No recognition default is changed by this package.

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1 Research objective and the audited baseline

The objective is an algorithm, starting from a classical knot diagram with n crossings, whose complete running time has a proved bound of $\exp((\log n)^{O(1)})$. Efficient arithmetic on a supplied normal surface is necessary, but a recognition algorithm must also find the right surfaces, justify every simplification, transport the required boundary data, and bound the total work over the entire hierarchy. We work at that interface between local compression and global topology.

The repository audit was made at commit `704600626606a9097cfd89d4e3d2bbc63c8dd1cb`, dated 12 October 2026. Both the maintained `Topology/UnknotRecognition` implementation and the four archives then present in `docs/incoming` were inspected [9]. The maintained recognizers describe a general exponential guarantee. They already contain extensive local compression machinery: weighted interval orbits, normal-cut cells, source-bound parallelity geometry, ordered boundary words, handle cancellation, cohomology transport, and independent certificate checkers. The present work therefore begins with their actual interfaces rather than proposing those ingredients afresh.

1.1 What is inherited, and what is added

The incoming material materially changes the research starting point. Table 1 records the distinction used throughout this article. Exact filenames, source paths and SHA-256 hashes appear in the accompanying `SOURCE_AUDIT.json`. These archives are project research documents, not external peer-reviewed results; a statement inherited from one is identified as such, and is not silently promoted to an implemented property of the maintained recognizer.

Incoming package, 12 October 2026	Relevant existing contribution	Consequence for this continuation
Moment-coded first returns	Abstract endpoint moments, exact edge mass, finite-group lifts, and passive component accounting	The missing source-to-owner bridge is implemented on native parallelity geometry; the pair encoding is sharpened.
Disconnected tile kernels	Compressed powers and disconnected-component handling	We do not claim a new arbitrary-power kernel. The unresolved native chart and hierarchy interfaces remain research tasks.
Unknot hierarchy kernels	Clustered search, interval bundles, and local hierarchy kernels	A small local kernel is not identified with a complete bounded-depth hierarchy.
Sparse cut grammars	Bounded-width compressed response constructions	Geometric realizability and width control must precede use in a global complexity proof.

Table 1: The four pending packages used to distinguish inherited work from the additions in this report.

There are two concrete gaps in the maintained code that motivate the new implementations. The source-handle reducer can expose one-handles that meet no remaining two-handle. Its existing positive certificate then transports a cocycle back to the original boundary to establish essentiality. Unused bridges do not supply that nonseparating witness, and the code explicitly leaves their ball sides as a next reduction. We process every unused handle together, use irreducibility only through an authenticated geometric contract, and retain the unique torus core when appropriate.

The maintained parallelity-boundary compiler obtains its cyclic interface from two weighted scalar censuses using endpoint masks. The incoming moment report proposes arithmetic labels abstractly, but does not supply an authenticated owner for every actual native old-boundary edge.

We construct those owners from the free-edge records, count each edge exactly once on each fibre sheet, and assemble the native cyclic output directly from one census. This supplies exact lengths in addition to the existing word and local fibre data.

1.2 Main results and their status

1. **A genus budget and a sharp attachment bound.** If M is compact, connected, orientable and irreducible, with connected boundary of genus g, cutting unused one-handle cocores gives pieces K_v and a connected attachment multigraph Q satisfying

$$g = \beta(Q) + \sum_{K_v \text{ nonball}} (1 - \chi(K_v)).$$

   Every nonball term is at least one. A rooted forest then absorbs all ball pieces, leaving either at most g specified nonball pieces joined by at most $g - 1$ handles, or one ball with g handles. The general topological theorem and its graph certificate are proved in Section 3. The graph implementation does not construct parametrizations of discarded balls.
2. **A complete initial empty-pattern reduction.** For an irreducible torus source the budget has exactly two cases. One certifies a solid torus; the other retains a unique Euler-zero component homeomorphic to the source. The deletion-only dual tetrahedral presentation gives an additional finite local-type theorem, making this retained core admissible for the initial empty pattern. The source compiler, independent replay and canonical-diagram wrapper are implemented in Section 4. Reducible sources with the same numerical signature are included as executable counterexamples to omitting the irreducibility hypothesis.
3. **A proved improvement in certificate arithmetic.** Cancellation can be replayed by decrementing nonnegative geometric occurrence counts, visiting each deleted attaching row once. On the specified Fibonacci layered-solid-torus family, the native explicit transported dual vector requires $\Theta(t^2)$ bits. The new incidence/component certificate has $O(t \log(t + 1))$ bits and avoids that transport. This is a comparison with the precise maintained format; it is not a lower bound for every possible compressed cocycle.
4. **A native one-census boundary operation.** Source ownership and a scalar moment encoding recover the full native cyclic skeleton, exact unmarked gap lengths, and local fibre flips. For p ports, terminal labels use $O(\log(p + 1))$ bits. Exact integer arithmetic and a global inventory check recover pairs without probabilistic hashing. Section 5 supplies the geometry, proof, independent source-bound verifier, comparisons, and limits of the output.
5. **An output-sensitive reconstruction statement.** For a supplied orientable normal surface S in a t-tetrahedron source, the entire unmarked cut manifold admits a triangulation of size $O(t + b_0(S) + \chi_-(S))$ under the explicit compressed-cut construction. This is derived from published parallelity machinery and surface classification. It counts all components and charges attaching data. It is a theoretical consequence here, not a delivered retriangulation implementation; see Section 6.
6. **A global accounting theorem and testable research targets.** Section 7 states sufficient quantitative conditions for the announced exponent, with the unresolved assumptions visible. Section 8 turns those assumptions into specific mathematical and implementation questions.

1.3 Conventions and epistemic boundaries

All integer input is binary. A normal vector with a B-bit entry is not expanded into its individual discs. t denotes the current number of tetrahedra, W a normal weight, and p the number of retained core ports. The native corner namespace has size N; N can be exponentially larger than its binary description. A compressed component type with multiplicity m is one record with a binary count, unless explicit expansion is part of the requested output. Every asymptotic

statement distinguishes arithmetic bit cost, incidence updates and output size where this affects the claim.

A valid decomposition of a supplied manifold and a recognition verdict are different outputs. Source validation checks that the finite triangulation or normal packet is the one being discussed. It does not turn a torus boundary into irreducibility. The diagram wrapper supplies the latter through a checked construction of a classical knot exterior in S^3. Likewise, an incidence hash identifies finite input data; it is not a proof that an abstract Euler-one label represents a ball.

The term *independent checker* means that a checker does not call its producer's discovery routine and reconstructs the relevant source through the stated separate path. It does not mean a formal proof assistant has verified the Python code, or that no shared parser or arithmetic library is used. Tests that deliberately disable producer routines make this software boundary observable. Exact proofs establish the mathematical statements; finite tests assess the implementations on their specified domains.

The new results are contributions relative to the inspected project state. Disc cutting, boundary connected sum, surface classification, moment identities and union–find are established ingredients. No general priority claim is made for these ingredients or for every possible formulation of the deductions. The claimed advances are the precise combined statements, their source-authenticated implementations, the quantitative separation on a stated family, and the research obligations that remain after them.

2 Sources and the announced quasipolynomial bound

The source record contains two different quantitative versions of Lackenby's 2021 announcement. They should not be silently identified.

Source	Stated running time	Stated hierarchy depth
February 2021 author slides	$n^{O(\log n)}$	$O(\log n)$
March 2021 Oxford slides	$2^{O((\log n)^3)}$	$O((\log n)^2)$
15 April 2021 Princeton abstract	$2^{O((\log n)^3)}$	not specified

Here n is the number of crossings of the input knot diagram. The first row is explicit in the February slides, including their complexity summary; it also agrees with Oxford's news announcement [3, 8]. The later slide version and Princeton abstract state the weaker, still quasipolynomial bound [4, 5]. Thus the earlier $n^{O(\log n)}$ figure is a documented historical claim, while

$$2^{O((\log n)^3)} = n^{O((\log n)^2)}$$

is the later explicit benchmark used here. Matching the stronger exponent would require an additional argument, not a change of notation.

The later slides describe a hierarchy procedure whose cost is controlled by its depth and a polynomially bounded surface parameter. They combine compressed parallelity bundles, multisurfaces, and modifications of generalised Heegaard splittings. Their logarithmic-squared depth bound uses a dichotomy involving boundary-genus growth and special regions satisfying a Cheeger-type inequality [4]. This architecture is more than an efficient enumeration of fundamental normal surfaces. A multisurface in those slides consists of individually embedded surfaces representing independent relative homology classes; the surfaces may intersect, and successive cuts are taken in their complements. A collection of simultaneously disjoint representatives cannot be substituted without proof.

As of this article's source audit on 12 October 2026, the sources located for the quasipolynomial complexity theorem are announcements and slides. A complete published proof of that bound was not located. This is a statement about the material found in the audit, not a claim that

no other manuscript exists. The author posted a substantial new preprint, *Incompressible surfaces, hierarchies and unknot recognition*, on 25 July 2026 [7]. It develops an algorithm for incompressibility and essential boundary patterns in irreducible compact orientable 3-manifolds, and discusses future speed-ups. Proposition 9.1 bounds an iteration count by $L(g+1)^L$ in terms of the maximum hierarchy length L and pattern complexity g; it does not supply the bounds on those parameters needed for the announced exponent. Theorem 14.4 provides compressed parallelity-bundle extraction for uniform handle structures. Its stronger admissibility procedure in Proposition 14.2 assumes both irreducibility and an essential boundary pattern. Those hypotheses must be checked before using that procedure in an unknot-recognition implementation.

Several necessary ingredients are already published. Agol, Hass, and Thurston give polynomial-time orbit counting for interval pairings and a weighted extension useful for normal-surface component data [1]. Their weighted output permits an interval of orbits with one common weight vector; multiplicities need not be expanded. This matters whenever the normal input contains exponentially many repeated components. Lackenby's certification paper supplies the explicit core and ordered interval-bundle attachments used in Section 6 [2]. These results justify compressed cutting, but do not establish a fast method of finding a suitable short hierarchy.

Boundary markings have their own algorithmic theory. The published *Some Fast Algorithms for Curves in Surfaces* gives algorithms for geometric intersection numbers and, for essential curves and arcs, isotopy, with polynomial dependence on triangulation size and logarithmic dependence on normal weights [6]. This offers a concrete source of subroutines for a future marking-aware implementation. It does not by itself bound the complexity of transferring an arbitrary hierarchy pattern through a 3-dimensional reconstruction. The research program must state that transfer problem separately and prove its complexity.

3 A genus budget and a small attachment network

The initial torus-boundary reduction is the smallest instance of a more general structural fact. Its useful parameter is the topology of the boundary, rather than the number of zero-handles in a presentation. We give the argument in a form that separates a topological theorem from the combinatorial certificate implemented in this package.

3.1 Cut pieces and the unused-handle graph

Let M be a connected compact orientable irreducible three-manifold with nonempty boundary. Suppose a finite handle presentation has been given and a set U of one-handles is disjoint from the attaching regions of all higher-index handles. Their cocores form disjoint properly embedded discs. Cut along all these discs. Let K_v, for $v \in V(Q)$, be the connected pieces, and put one edge of Q for every original handle in U, with endpoints its two incident pieces. Loops and parallel edges are retained. The graph Q is connected, and

$$q = |V(Q)|, \qquad e = |E(Q)|, \qquad \beta = e - q + 1.$$

The index β is a graph cycle rank, not the first Betti number of the whole manifold. Write

$$\delta(M) = 1 - \chi(M), \qquad d_v = 1 - \chi(K_v).$$

When ∂M is connected of genus g, orientability and Poincaré–Lefschetz duality give $\delta(M) = g$. For boundary components of genera $g_1, \dots, g_b$, the corresponding formula is $\delta(M) = 1 - b + \sum_i g_i$.

Lemma 3.1 (Irreducibility and Euler labels after disc cutting)**.** *Every K_v is irreducible and has nonempty boundary. It is a ball if and only if $\chi(K_v) = 1$. Every other piece satisfies $\chi(K_v) \le 0$.*

Proof. An interior sphere in a cut piece is also an interior sphere in M, so it bounds a ball in M. The cutting discs are disjoint from the sphere. Each disc is connected and has boundary on ∂M, outside that interior ball. It therefore lies outside the ball. Consequently the ball survives inside the same cut piece. Disc cutting cannot create a closed component, which proves the first assertions.

A connected irreducible three-manifold with a sphere boundary component is a ball: push that component into the interior and apply irreducibility; the side away from the original boundary is a ball, and restoring the spherical collar gives a ball. Thus a nonball K_v has only boundary components of genus at least one. The identity $2\chi(K_v) = \chi(\partial K_v)$ now implies $\chi(K_v) \leq 0$. Conversely a ball has Euler characteristic one, and the two possibilities exhaust the pieces. □

This lemma uses actual irreducibility of the source. An abstract integer label equal to one, a homology computation, or a sphere boundary alone does not establish the conclusion. The executable reducible controls in Section 9 show why the distinction matters.

Theorem 3.2 (Exact unused-handle budget)**.** *Under the preceding hypotheses,*

$$\boxed{\delta(M) = \beta + \sum_{v \in V(Q)} d_v = \beta + \sum_{K_v \ nonball} (1 - \chi(K_v)).} \tag{1}$$

In particular, if s is the number of nonball pieces, then $s + \beta \leq \delta(M)$. For connected boundary this is $s + \beta \leq g$.

Proof. Recover M by attaching the e one-handles to the disjoint union of the K_v. Each such attachment decreases Euler characteristic by one, so $\chi(M) = \sum_v \chi(K_v) - e$. Substituting $e = \beta + q - 1$ gives

$$1 - \chi(M) = \beta + \sum_v (1 - \chi(K_v)).$$

By Lemma 3.1, a ball contributes zero and every nonball contributes an integer at least one. This proves the equality and bound. □

The identity does more than count components: it distinguishes the two possible sources of boundary complexity, cycles in the attachment graph and topological complexity inside the retained pieces. Neither term can be large while the boundary genus remains small.

3.2 Absorbing ball pieces along a rooted forest

For the following topological replacement, a network means the graph together with its vertex manifolds and its labelled attaching discs. The graph without these interfaces is not a complete manifold encoding. No boundary pattern or prescribed embedded curve is part of the replacement contract.

Lemma 3.3 (Absorption of a ball vertex)**.** *Suppose a ball vertex B is joined to a different vertex manifold K by a chosen one-handle. The union of K, this handle, and B is homeomorphic to K. Under such a homeomorphism all other attaching discs on B can be moved to mutually disjoint subdiscs of a neighbourhood of the chosen attaching disc on ∂K. The homeomorphism can be taken to fix the complement of a chosen three-dimensional boundary collar in K.*

Proof. A ball attached to one end of a one-handle is a ball with a distinguished disc at the other end. Gluing this ball to K along that disc adds a boundary collar. Removing the collar gives the asserted homeomorphism. The remaining attaching discs are a finite labelled collection of disjoint discs on the boundary of the added ball, away from its distinguished disc. A homeomorphism of the boundary sphere carries them into any prescribed disjoint subdiscs in the corresponding

collar patch. It extends over the ball. This argument preserves labels and orientation, including when some of the other handles have both ends on B. It does not preserve an arbitrary additional boundary pattern. □

Theorem 3.4 (Genus-controlled attachment kernel)**.** *Assume the hypotheses of Theorem 3.2, and suppose the attaching discs have been supplied. There is a topologically equivalent unmarked attachment network with the following sizes.*

(i) If $s \geq 1$, the vertex manifolds are exactly the s nonball pieces, and the number of one-handles is

$$e' = \beta + s - 1 \leq \delta(M) - 1. \tag{2}$$

Thus the network has at most g vertex manifolds and $g-1$ handles when the original boundary is connected of genus g.

(ii) If $s = 0$, the network consists of one ball and $\beta = \delta(M)$ one-handles. In particular M is a handlebody of genus $\delta(M)$.

The source-to-kernel combinatorial correspondence has a rooted-forest certificate of $O(q+e)$ records, constructible and checkable in polynomial bit time. This is a bound for the attachment network; it is not a bound for the internal presentation size of the retained nonball pieces.

Proof. If $s \geq 1$, choose a spanning forest F of Q with exactly one nonball vertex in each tree. One constructive method starts with all vertices separate, and processes the edges, joining two distinct components unless both already contain a nonball vertex. At most one root is ever present in a component. At termination, any component with no root would have an edge to another component because Q is connected. That edge could not have been rejected by the rule. Hence every final component has a root and there are exactly s trees.

Root each tree at its nonball vertex. Starting at the leaves, absorb every other vertex and its parent handle using Lemma 3.3. Every absorbed vertex is a ball. Incidences of handles outside F are retained, with their attaching discs carried into the appropriate parent patch. Exactly $q - s$ handles are removed, leaving $e - (q - s) = \beta + s - 1$. The bound follows from Theorem 3.2.

If every piece is a ball, choose one vertex as the root of a spanning tree and carry out the same procedure. The remaining ball has $e - (q - 1) = \beta$ one-handles, the defining presentation of an orientable handlebody of genus β. Equation (1) gives $\beta = \delta(M)$.

For the combinatorial certificate retain the original edge identifiers, the edges of F, the root of every source vertex, and one parent pointer per nonroot vertex. Every surviving handle is still an original nonforest edge. Its new ends are the roots of its old ends. A checker verifies the source hash, all forest edges, acyclicity, unique roots, coverage of every source vertex, every remaining incidence, and the Euler and cycle-rank equalities. Sorting and union–find give $O((q+e)\log(q+e))$ comparison work, with binary costs charged separately. The independent checker can traverse the supplied rooted trees directly. □

3.3 Source addresses and the meaning of a small kernel

An endpoint originally on an absorbed vertex is rerouted along its parent chain. These chains must be shared. Explicitly writing the complete route for every endpoint of a long chain can require quadratic space; one parent pointer per source vertex suffices. The implementation returns this forest. After a one-time $O(q)$ parent-index construction, it permits expansion of one requested route in time proportional to its length. The convenience API pays that indexing cost if no index is supplied. It never expands every route as part of kernel construction or verification.

There are three distinct outputs here. The operational graph has the small size in Theorem 3.4. Its source provenance still has $O(q + e)$ records. An explicit homeomorphism from a triangulated ball piece to a standard ball is a further geometric output and is not constructed by the graph

routine. The topological replacement is justified by the preceding lemmas once the vertex pieces and source irreducibility are authenticated. A later patterned-hierarchy consumer must additionally construct and verify the required disc charts and pattern transport. The initial torus compiler avoids this extra task by retaining its unique core subpresentation directly.

Proposition 3.5 (Sharp counts for this replacement contract)**.** *The bounds $s \le g$ and $e' \le g-1$ in the nonball case are attained simultaneously for every $g \ge 1$. In the all-ball case the one-vertex graph requires exactly g loops.*

Proof. Take g solid tori and connect them by $g-1$ one-handles in a tree. The result is a handlebody of genus g, hence irreducible. Regard the connecting handles as U. Its pieces are the g solid tori, each with Euler characteristic zero, so $s = g$ and $\beta = 0$. Arbitrarily many ball vertices can be inserted along the tree without changing the example. All g specified nonball vertex manifolds remain under the stated replacement contract, and connecting them requires at least $g-1$ handles. If every piece is a ball, cycle rank is g and a one-vertex graph has exactly one independent cycle per loop. □

This sharpness is relative to retaining the specified nonball blocks. A handlebody could of course be given another global presentation. More generally, a knot exterior of arbitrarily high presentation complexity can be the sole nonball vertex when $g = 1$. A one-vertex attachment graph therefore gives no uniform bound on the difficulty of recognizing that vertex manifold. This is why the result improves a hierarchy primitive without proving general quasi-polynomial unknot recognition.

3.4 Implementation and independent graph replay

`attachment_kernel.py` implements the rooted-forest producer. `attachment_kernel_verify.py` imports no producer code and checks the complete finite certificate. The adapter `kernel_from_decomposition` converts the decomposition format from `unused_handles.py`; it does not call that module's geometric verifier. The caller must first authenticate the decomposition against the original source. In particular, the graph certificate's hash binds only the vertex Euler labels and edge incidences. It does not bind a triangulation of each vertex manifold or its attaching-disc charts. A topological consumer must retain the separate verified source packet and, when invoking ball absorption, its irreducibility justification. Loops, parallel handles, isolated one-vertex networks, zero genus and nonball deficits larger than one are part of the supported graph model. The tests include every connected labelled simple graph on at most five vertices under three root assignments, random multigraphs, sharp families, and altered proof records.

4 Unused cocores, a genus budget, and initial admissibility

The maintained initial handle operator removes geometric two/three-handle pairs, removes geometric one/two-handle pairs, and tests exposed cocores by transporting a dual integral cocycle to the original boundary. Its output explicitly stops short of an admissibility assertion: an unused one-handle has two characteristic zero-handles of index -4. The final paragraph of the maintained `source_handles.tex` identifies unused bridges and their ball sides as a next operation. We resolve that initial, empty-pattern operation simultaneously for all unused handles. No later boundary pattern is discarded.

The topological ingredients—disc cutting, irreducibility, and boundary connected sum—are classical. The contribution here is an exact budget for their source-handle quotient, an authenticated initial compiler, a smaller certificate on a specified family, and a faster independent replay of the supplied cancellation trace. This is an explicit specialization of Lackenby's admissibility procedure, not a claim to invent admissibility or to complete the general patterned hierarchy.

4.1 The geometric quotient

Let M be a compact connected orientable three-manifold with nonempty boundary. Consider a finite handle presentation containing zero-, one-, and two-handles, with all three-handles already removed. For a one-handle e, let $d(e)$ be the total number of its intersections with all two-handle attaching curves, counted as geometric occurrences. Oppositely oriented passages both contribute one. Call e *unused* if $d(e) = 0$ and *used* otherwise.

The graph of zero- and one-handles has one vertex for each zero-handle and one edge for each one-handle, with loops and parallel edges retained. Delete its unused edges. For each resulting connected vertex block, take the zero-handles in that block, its used one-handles, and all two-handles attached there. Write the resulting connected three-manifolds, after rounding corners, as $K_1, \dots, K_r$. Define Q to have these r pieces as vertices and the unused one-handles as edges. Thus an edge of Q retains its source one-handle identifier and its two oriented endpoint occurrences. Put

$$\beta = |E(Q)| - |V(Q)| + 1.$$

Lemma 4.1 (Exact unused-cocore decomposition)**.** *Every two-handle belongs to exactly one K_i. The graph Q is connected, and cutting M simultaneously along the cocores of all unused one-handles gives precisely the pieces K_i. In particular,*

$$\chi(M) = \sum_{i=1}^{r} \chi(K_i) - |E(Q)|. \tag{3}$$

Moreover, with arbitrary choices of connecting paths,

$$\pi_1(M) \cong \left(\mathop{*}_{i=1}^{r} \pi_1(K_i) \right) * F_\beta. \tag{4}$$

Proof. An attaching circle cannot pass through an unused handle. Its successive zero-handle passages are therefore connected by used edges and belong to one block. This gives a well-defined, complete assignment of two-handles. The cocore of an unused one-handle is a properly embedded disc because its boundary misses all two-handle attaching regions. The chosen cocores are disjoint. Cutting each product $D^2 \times I$ across $D^2 \times \{1/2\}$ and absorbing the two half-products into its endpoint zero-handles removes the corresponding graph edge and no used handle. Thus the connected pieces are exactly those defined above.

The original manifold is connected and adding handles of index two does not connect graph components, so Q is connected. Reattaching one unused one-handle decreases Euler characteristic by one, which proves (3). First attach a spanning tree of Q. Each edge joins two components along discs, and van Kampen gives their free product. Each of the remaining β one-handles adds one free generator and no relation. This proves (4). □

The quotient uses actual attaching occurrences. Replacing an attaching word by its abelianized coefficient row before deciding whether a handle is unused invalidates the lemma. In particular, exponent sum zero is not geometric disjointness.

4.2 The genus budget in a torus source

Lemma 3.1 shows that, under authenticated irreducibility, the K_i are irreducible, Euler characteristic one identifies precisely the ball pieces, and every nonball piece has nonpositive Euler characteristic. Let $\mathcal{N}$ index the nonball pieces. Theorem 3.2 specializes to

$$g = \beta + \sum_{i \in \mathcal{N}} (1 - \chi(K_i)), \qquad \beta + |\mathcal{N}| \le g, \tag{5}$$

when the source boundary is connected of genus g. The number of ball pieces is absent from this identity. Its torus case has a particularly strong consequence for the initial source operator.

Corollary 4.2 (Torus dichotomy)**.** *For an irreducible M with one torus boundary, exactly one of the following numerical cases occurs.*

1. *$\beta = 1$ and every K_i is a ball. Then M is a solid torus.*
2. *$\beta = 0$, exactly one K_i has Euler characteristic zero, and all other pieces are balls. The graph Q is a tree, the exceptional piece has one torus boundary, and it is homeomorphic to M.*

The second case is compatible with both trivial and nontrivial knots.

Proof. Every term of (5) is a nonnegative integer, and the total is one. If $\beta = 1$, there is no nonball piece. Attach a spanning tree of Q to the balls; the result is a ball. Its single remaining one-handle makes a solid torus.

Otherwise $\beta = 0$ and there is exactly one nonball piece, whose Euler characteristic is zero. A connected graph of cycle rank zero is a tree. Each reattached tree edge joins different manifold components and hence joins two different boundary components. If b_i is the number of boundary components of K_i, then

$$1 = \sum_i b_i - (r - 1).$$

Each $b_i \geq 1$, so every $b_i = 1$. The nonball piece therefore has a connected boundary of genus one. A boundary connected sum with a ball does not change a manifold's homeomorphism type. Absorbing the ball pieces along the tree proves the claim. □

4.3 The complete initial empty-pattern compiler

For a triangulated source, the maintained construction has $O(t)$ handles and $O(t)$ attaching occurrences, where t is the number of tetrahedra. The first queue removes a two-handle meeting exactly one surviving three-handle, with endpoint multiplicity retained. Every source interior vertex is connected to the boundary in the one-skeleton, so this phase removes all three-handles. The second queue removes a one-handle meeting exactly one two-handle in exactly one disc, counted across *all* surviving two-handles. No surviving two-handle passes through the cancelled one-handle, so no implicit handle slides or relator substitutions are used. These are precisely the geometric prerequisites of the maintained source-handle cancellation trace.

At the fixed point, every surviving one-handle has degree zero or at least two. Applying Corollary 4.2 gives the following result.

Lemma 4.3 (The local tetrahedral characteristic graph)**.** *After the stated deletion-only reductions and removal of unused handles, the characteristic graph in each zero-handle of the retained torus core is a nonempty connected simple subgraph of K_4. Its regular neighbourhood is the characteristic surface, and every complementary region in the zero-handle boundary is a disc. Up to tetrahedral symmetries, its four possible graph types are C_3, C_4, $K_4 - e$, and K_4. Every characteristic zero-handle has index zero or two; the total indices in these four local types are respectively zero, zero, four, and eight.*

Proof. The vertices of the local characteristic graph are the surviving formal facet ports of the source tetrahedron. There are at most four. A surviving characteristic strip corresponds to an original local tetrahedral edge and joins its two distinct incident facets. There is at most one such local edge for each unordered pair of facets. Thus the graph is simple, even if different local vertices or strips have the same *global* handle identifier. The recorded reductions only delete whole handles; they do not identify local ports or insert new characteristic strips.

The degree at either local end of a surviving one-handle e is exactly its geometric attaching-occurrence count $d(e)$. This count is at least two after the degree-one cancellations and unused-handle removal. Every nonempty connected component of a simple graph of minimum degree at

least two has at least three vertices. Two such components cannot fit on at most four vertices. Hence the local graph is connected whenever nonempty.

A zero-handle in the retained used-edge component has a used port unless it is the sole zero-handle of an isolated component. In that exceptional case no source two-handle can attach there without a one-handle passage, and its component is a ball of Euler characteristic one, so it is not retained. Thus every retained local graph is nonempty. The regular neighbourhood of a connected graph embedded in a sphere is a connected planar surface, and all its complementary components are discs.

For completeness, three local vertices force the triangle C_3. With four vertices, every degree is two or three. The graph has four, five, or six edges. The four-edge case is C_4, since every degree is two; the five-edge case deletes exactly one edge from K_4; and the six-edge case is K_4. These are the four types listed, with their embeddings in the tetrahedral sphere inherited from the source. A local port of degree d has index $-4 + 2d$, which proves the stated index bounds and totals. □

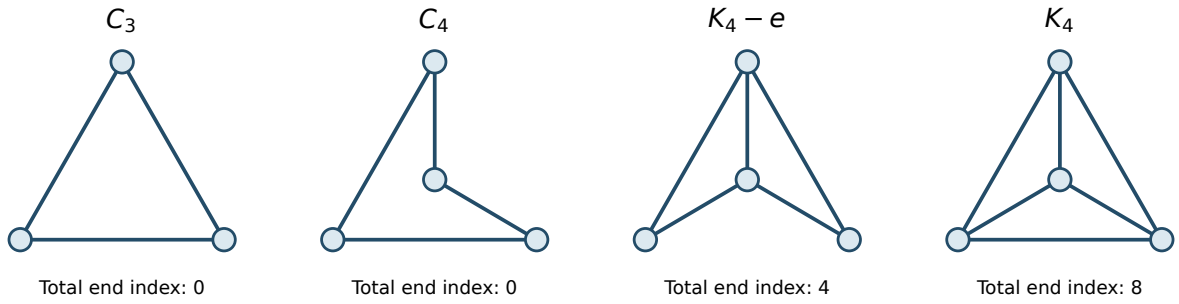


Figure 1: The four possible nonempty local characteristic graphs in the retained initial core. Nodes are formal local face-port discs; edges are surviving two-handle strips. Every end has index zero or two. The drawing is a combinatorial schematic; the actual source embedding is retained in the certificate.

Theorem 4.4 (Initial admissibility with authenticated irreducibility)**.** *Given a finite triangulation of a compact connected orientable irreducible three-manifold with one torus boundary and empty boundary pattern, the incidence cancellations and unused-cocore quotient give, in polynomial encoded time, either*

1. *a solid-torus certificate from the first case of Corollary 4.2; or*
2. *an inherited handle presentation, homeomorphic to the source, that is admissible for the empty pattern.*

The second output consists of the handles in the unique Euler-zero piece; the source charts and cancellation trace retain their provenance. No normal surface search or boundary-period transport is required.

Proof. Termination and polynomial cost of the cancellation queues follow because each move removes a handle pair and each attaching occurrence is removed at most once. The quotient and Euler counts are finite graph operations. In the cycle case the preceding corollary gives the solid torus.

In the tree case cut all unused cocores and retain the unique Euler-zero piece. The other pieces are balls, and the corollary gives a homeomorphism with the original manifold. Every one-handle of the retained presentation is used and hence has $d(e) \geq 2$. Each of its characteristic zero-handle ends has index

$$-4 + 2d(e) \geq 0.$$

An index-zero end is disjoint from the pattern, because the pattern is empty. For the second condition, Lemma 4.3 shows that every complementary region of the characteristic surface in a zero-handle boundary is a disc. A simple closed curve disjoint from the characteristic surface therefore bounds a disc in that same complement, and the disc meets the empty pattern in the empty set. This proves the stronger complement-disc condition, as well as the printed admissibility condition; we do not rely on the weaker observation that an arbitrary sphere curve bounds a disc that might contain characteristic handles. Both required conditions therefore hold. □

The retained core is a handle subpresentation, not a fresh triangulation. It includes inherited source identifiers, its assigned two-handles, and the geometric reduction trace. This suffices to authenticate its unmarked homeomorphism type and initial admissibility. It does not provide an explicit parametrization of every discarded ball, a transport map for an arbitrary later pattern, or a normal-coordinate adapter on a retriangulated core. Those remain implementation tasks.

For an input knot diagram, the implementation's wrapper independently checks that the supplied triangulation is the canonical exterior of that exact classical one-component diagram. A knot exterior in the three-sphere is irreducible: a sphere disjoint from the connected knot bounds a ball on the side not containing the knot. Thus the wrapper authenticates the theorem's hypothesis. It returns `UNKNOT` in the cycle case and `ADMISSIBLE_CORE` in the tree case. The latter is an intermediate result, never a knottedness certificate.

4.4 Why the irreducibility gate cannot be replaced by a flag

The delivered exact controls `solid_torus_sum_rp3` and `solid_torus_sum_s2xs1`, copied from the pinned source corpus, are reducible manifolds with torus boundary. Both produce $\beta = 1$ with Euler characteristic one for every cut piece. An independent Regina check finds that neither is a solid torus. In particular, the quotient and the Euler data alone do not distinguish the ball pieces from punctured closed summands.

This has a direct classical model. Let N be a closed orientable three-manifold other than the sphere, remove a ball, and take the boundary connected sum of the resulting punctured manifold with a solid torus. The result is the connected sum of the solid torus and N. Cutting a meridian cocore leaves a piece with Euler characteristic one, but the punctured N remains. If $N = \mathbb{RP}^3$, its fundamental group already precludes a ball. If $N = S^2 \times S^1$, the additional infinite-cyclic free factor does the same.

The generic `verify_unused_handle_certificate` therefore authenticates only the geometric decomposition and its numerical case. It never authorizes an irreducibility assertion from a caller's Boolean. The diagram wrapper verifies exterior provenance first. Mutating a valid diagram certificate to substitute either reducible source is rejected.

4.5 Linear replay of geometric cancellation traces

The maintained independent cancellation checker reconstructs source cells separately from the producer. For each proposed one/two-handle cancellation it scans every surviving two-handle attaching cycle to confirm that the selected one-handle occurs once in total. This is sound but can repeatedly scan the same surviving rows.

Proposition 4.5 (Occurrence-count replay)**.** *After independently reconstructing the geometric attaching rows, a supplied cancellation trace with h handles and L attaching occurrences can be checked using $O(h + L)$ combinatorial incidence updates, excluding ordered output construction and source reconstruction. Its acceptance condition for each one/two cancellation is identical to the full-scan condition.*

Proof. Let m_{ef} be the nonnegative number of occurrences of one-handle f in the attaching circle of two-handle e. Initialize $d_f = \sum_{e \text{ alive}} m_{ef}$. Deleting e subtracts m_{ef} from d_f for every nonzero entry of that row. At a proposed cancellation of f with e, require both to be alive and require

$$d_f = 1, \qquad m_{ef} = 1.$$

Since all multiplicities are nonnegative, these conditions say exactly that e supplies the unique geometric occurrence of f among all alive two-handles. Signed cancellation plays no role. The two/three-handle condition is checked separately using the surviving endpoint occurrences, and its deleted row is charged in exactly the same way.

Each row is deleted at most once. Its incidence entries are visited once at initialization and once at deletion. The handle-state tests and final counts use linear additional work. An independently recomputed final degree table can be checked in one further pass without changing the bound. □

The prototype uses union by size and path compression for the producer's quotient, followed by canonical component numbering. The verifier uses breadth-first search on separately reconstructed raw source cells, without calling the producer, its union-find routine, or its scheduler. It also checks the local facet ports and source-edge strips in Lemma 4.3, including their geometric degrees and local connectedness. The two paths share immutable input formats and elementary arithmetic; the source geometry constructions are the established distinct maintained paths.

All source identifiers, degrees, and component sizes have $O(\log(t+1))$ bits. The incidence and quotient certificates contain $O(t)$ records and therefore occupy $O(t\log(t+1))$ bits. Original source validation, fingerprinting, and raw-cell reconstruction remain additional polynomial costs. The linear incidence theorem does not assert that the entire Python source-validation stack has linear bit complexity.

4.6 A proved certificate-size separation on Fibonacci layered tori

The standard layered-solid-torus fixture makes the eliminated boundary arithmetic explicit. Number its tetrahedra $0, \dots, t-1$. For $i < t-1$, face zero of tetrahedron i is paired to face two of $i+1$ by the vertex permutation $(2,1,3,0)$, and face one is paired to face three of $i+1$ by $(0,3,1,2)$. In the last tetrahedron, face zero is paired to face one by $(1,2,3,0)$. Reverse pairings are the inverse permutations; faces two and three of tetrahedron zero remain boundary. This is the maintained Fibonacci layered-solid-torus family, not a newly proposed topological construction.

For $0 \le i \le t-2$, write a_i, b_i for the dual one-handles of face zero and face one of tetrahedron i, and write c for the final self-paired handle. Their native identifiers are $2i, 2i+1, 2t-2$, respectively. Following the interior-edge links through the displayed face permutations gives the attaching words

$$r_i = b_i a_{i+1} a_{i+2} b_{i+2}^{-1} b_{i+1}^{-1} a_i^{-1} \quad (0 \le i \le t-4),\ r_{t-3} = b_{t-3} a_{t-2} c\, b_{t-2}^{-1} a_{t-3}^{-1},\ r_{t-2} = b_{t-2} c^2 a_{t-2}^{-1}. \quad (6)$$

The penultimate line is present for $t \ge 3$, the last for $t \ge 2$; the case $t = 1$ has only c and no two-handle. The first line is the repeating six-sector link. The final two lines are its five-sector and four-sector closures at the self-paired end. Here is an explicit check of the three local types. Write $(j, uv; p, q)$ for the sector at edge uv in tetrahedron j, entering through facet p and leaving through facet q. For $0 \le i \le t-4$ the successive sectors are

$$\begin{aligned}(i, 23; 0, 1) &\longrightarrow (i+1, 12; 3, 0) \longrightarrow (i+2, 13; 2, 0)\\ &\longrightarrow (i+3, 01; 2, 3) \longrightarrow (i+2, 02; 1, 3) \longrightarrow (i+1, 03; 1, 2).\end{aligned}$$

The last exit returns to the first entry. At $i = t-3$, the self-pairing replaces the middle step through tetrahedron $i+3$ by a direct passage from $(i+2, 13; 2, 0)$ to $(i+2, 02; 1, 3)$, giving the five-sector word. At $i = t-2$, the four sectors are

$$(i, 23; 0, 1) \longrightarrow (i+1, 12; 3, 0) \longrightarrow (i+1, 23; 1, 0) \longrightarrow (i+1, 03; 1, 2).$$

Reading the exiting facet pairings gives exactly (6), including two successive positive crossings of c in the last case. The calculation uses only the displayed face permutations, so it proves the words for every t.

Proposition 4.6 (Fibonacci transport and certificate separation)**.** *For this family, the maintained queue cancels (a_i, r_i) in the order $i = 0, \dots, t-2$. Its remaining one-handles are $b_0, \dots, b_{t-2}, c$. The native transported cocycle for the surviving cocore c has*

$$x(b_i) = 0, \qquad x(c) = 1, \qquad x(a_i) = F_{t-i+1}. \tag{7}$$

Consequently, the native explicit dual-value list alone occupies $\Theta(t^2)$ binary bits. The new source incidence/component certificate occupies $O(t \log(t+1))$ bits and contains no transported cocycle.

Proof. Initially a_0 occurs exactly once. After rows $r_0, \dots, r_{i-1}$ are removed, a_i occurs exactly once in r_i. The native queue places a_i before b_i: in the row whose removal first makes them unique, the occurrence of a_i precedes that of b_i. Once a_i is cancelled, the pending b_i has degree zero and is skipped. Future a_j have more than one occurrence until this step, and c retains its double occurrence in the final row. This proves the stated cancellation order and remainder.

The remaining b_i form a tree joining the t zero-handles, while c is its single additional loop. The surviving-cocore assignment sets all b_i to zero and c to one. Reading the final two equations in (6) gives

$$x(a_{t-2}) = 2, \qquad x(a_{t-3}) = 3.$$

Every preceding row gives

$$x(a_i) = x(a_{i+1}) + x(a_{i+2}).$$

Uniqueness of reverse substitution proves (7), including the two small end cases.

The t nonzero values are $F_2, F_3, \dots, F_{t+1}$. The recurrence gives $F_{j+2} \geq 2F_j$ and $F_j \leq 2^{j-1}$, so $\log_2 F_j = \Theta(j)$. Summing their individual bit lengths proves the quadratic bound. In the new quotient every K_i is just one zero-handle, Q has t edges and t vertices, and the trace has $t-1$ cancellations. All identifiers are bounded by a constant multiple of t, proving the new upper bound. □

This is a comparison with the maintained explicit coordinate-list certificate, not a lower bound on every possible cocycle encoding. A recurrence-based encoding could also compress those Fibonacci integers. The new certificate avoids constructing and checking them at all because the authenticated geometric situation supplies a different proof. For general input diagrams the second branch of Theorem 4.4 may remain, so this size separation does not prove completeness of a polynomial unknot-recognition method.

4.7 Finite verification and performance scope

The delivered tests replay all 48 supplied source triangulations and compare the new cancellation sequence, remaining handle sets, summary, and source fingerprint with the maintained implementation. The same incidence-only certificates also pass the unchanged maintained cancellation verifier. Additional checks cover 48 independent local/global relabellings, canonical diagram exteriors, forged component lists and cycle ranks, Boolean labels, wrong diagram provenance, caps, cooperative interruption, and the two reducible controls. The replay tests disable the producer routines to confirm the independent call path. An optional Regina oracle supplies separate solid-torus and nonball checks.

There is also an unbounded geometric ballast family. Attach a chain of tetrahedral balls to embedded boundary triangles of the canonical trefoil exterior. The first triangle has three distinct quotient vertices, and each next one contains two old vertices and the new vertex, so

each attachment is an elementary boundary shelling. A reverse-shelling checker authenticates the family. Each extra tetrahedron contributes one ball component and one unused bridge, while the retained torus core is unchanged. The source packet therefore removes arbitrarily many irrelevant handles without any claim that the irreducible core itself has bounded complexity.

Source	t	Native ms	New ms	Native/New	Core h_0
Fibonacci $t = 1$	1	0.473	0.458	1.034	—
Fibonacci $t = 16$	16	6.250	6.180	1.011	—
Fibonacci $t = 64$	64	22.753	21.435	1.061	—
Fibonacci $t = 256$	256	173.290	159.682	1.085	—
Fibonacci $t = 512$	512	667.727	578.314	1.155	—
Canonical unknot	40	11.925	11.690	1.020	—
Canonical trefoil	60	19.005	20.213	0.940	48
Canonical figure-eight	80	29.950	28.445	1.053	68
Trefoil + 32 buds	92	30.324	29.663	1.022	48
Trefoil + 256 buds	316	108.484	117.183	0.926	48

Table 2: 5 paired repetitions per source, in randomized order. Values are medians of complete producer-plus-replay times. Ratios below one are regressions. A dash marks the all-ball cycle branch.

Fibonacci t	Native bytes	New bytes	Max. dual bits	Sum dual bits
1	581	699	1	1
16	2441	4010	11	111
64	9787	14865	44	1510
256	58924	59551	178	23103
512	172549	119503	355	91702

Table 3: Exact compact-JSON sizes and bit counts. The new decomposition has additional per-component records, so its asymptotic improvement need not appear at small sizes. Total dual bits includes one bit for each zero entry.

The canonical trefoil core retains $(h_0, h_1, h_2) = (48, 82, 34)$ from a source with 60 tetrahedra; the figure-eight core retains $(68, 118, 50)$ from 80 tetrahedra. Every added boundary bud leaves the trefoil core unchanged. At $t = 512$, the new Fibonacci certificate uses 119,503 bytes versus 172,549 bytes for the maintained certificate; at $t = 64$, it instead uses 14,865 versus 9,787 bytes. These finite measurements agree with the scope, and the crossover limitations, of Proposition 4.6. Timing regressions on some sources remain visible; the source-validation and raw-cell construction costs are included in both arms.

The benchmark measures completed source producer plus independent replay, with randomized paired order and all raw repetitions retained. It excludes serialization and canonical-diagram construction. The old query emits the geometric reduction and, on a hit, its transported boundary period; the new query emits the geometric reduction and the exact cut decomposition. Their source outputs differ, and the new topological interpretation uses separately authenticated irreducibility. These are source-operator measurements, not whole-recognizer timings. The maintained recognition defaults are unchanged and no new knot coverage is asserted.

5 A source-authenticated native moment compiler

The incoming moment construction left its passage from the maintained parallelity geometry to a weighted first-return query as a separate task. We implement that passage here. The resulting operator starts with an actual triangulation and binary normal coordinates, or with the complete parallelity certificate for that source. It produces the same cyclic core-port words and

attachment components as the maintained native boundary compiler, and adds the exact number of original-boundary edges in every complementary gap.

The maintained operator uses two scalar weighted censuses: one to identify marked corners, and one on the fibre double cover to identify residual gaps and their fibre transport. The new operator performs one scalar census on that same residual double cover. It replaces the corner query by canonically owned edge mass, and replaces the lifted endpoint masks by quadratic moments. These are improvements to an existing geometric operation. They do not supply a new cutting surface or establish a uniform bound on the number of stages in a recognition hierarchy.

5.1 The actual native packet

Fix a source with T tetrahedra and an accepted normal-coordinate vector. Let G be its compressed parallelity geometry at the audited source revision. A polygon family P has a binary rank interval and a distinct corner-occurrence interval for every corner of its polygon template. Write $c_{P,j}$ for the start of the interval at corner j. The occurrence of corner j at rank k has coordinate

$$x(P, j, k) = c_{P,j} + k - 1.$$

The entire corner namespace is $X = \{0, \dots, N-1\}$. Its size N can be exponential in the bit length of a normal multiplicity. No construction below enumerates X.

The packet contains a list of polygon-edge gluings. Such a gluing has affine rank maps of slope 1 or -1, an endpoint-reversal flag, and a separate fibre-reversal flag. The latter cannot be recovered from the slope of a singleton rank interval. The packet also gives a disjoint inventory of unglued polygon-edge occurrences. Each inventory record is either an original-boundary family or one singleton core-facing edge, called a *port*. Ports have distinct identifiers $0, \dots, p-1$ and the supplied ordering of their two local endpoints. The source construction has $p \le 66T$.

The geometric correctness of the source packet entails that its free edges, after corner identifications, form cellulated boundary circles. A circle may have only one edge; its two incidence flags are still distinct. The fibre labels define a genuine interval bundle over these circles and their corner stars. The reference source reconstruction in the checker is independent of the production geometry builder. Consequently the new checker does not accept a purported degree-two boundary merely because its moment histogram happens to look plausible.

Lemma 5.1 (Native canonical ownership)**.** *For an original-boundary record on edge j of polygon family P, with rank interval $[u, v)$, place unit edge mass on the interval*

$$[c_{P,j} + u - 1, c_{P,j} + v - 1) \subset X.$$

This assigns every original-boundary edge exactly one owner occurrence, and the owner represents one endpoint of that edge. The owner intervals of distinct original-boundary records are disjoint in X. In particular, if E is the represented number of original-boundary edges, then $E + p \le N$. For a original-boundary records the ownership program has a intervals, irrespective of the binary lengths of those intervals.

Proof. At fixed P, j, the inventory records are the disjoint rank gaps remaining after the explicitly recorded edge gluings; they cannot contain the same rank twice. The map $k \mapsto c_{P,j} + k - 1$ is injective. Different corner slots and different polygon families have disjoint occurrence namespaces by the construction of the $c_{P,j}$. Thus all the indicated intervals are disjoint. Their lengths are precisely the numbers of represented edge occurrences. Each recorded free edge belongs to this inventory once, so summing the lengths counts each original-boundary edge exactly once. The chosen corner is its local initial endpoint. Identifying corner aliases later does not change this count: several owners can become one actual vertex, and their masses are added there. Apply the same outgoing-corner assignment also to the p port singletons. They remain disjoint from

each other and from the original-boundary intervals, because the complete free-edge inventory is disjoint. These intervals have total length $E + p$ and lie in a namespace of size N, proving $E + p \leq N$. □

5.2 Compiling the residual fibre cover

Delete all open port edges. The remaining boundary is a union of compact paths, passive circles, and possibly silent occurrence classes that never meet a free edge. The source relation for the residual paths is generated by the corner gluings and by the endpoint identifications of all original-boundary edges.

To retain local fibre transport, replace X by $\widetilde{X} = X \times \{0, 1\}$, encoded as $x + sN$. A source corner gluing with fibre flag f joins sheet s to sheet $s \oplus f$. An original-boundary edge joins its endpoints on the same local sheet. Every source interval pairing therefore creates a bounded number of affine interval pairings in $\widetilde{X}$.

Lemma 5.2 (Residual degree and exact length)**.** *If $p > 0$, the residual components meeting port endpoints are exactly p paths, counting a point as a path of length zero. Each has two distinct endpoint occurrences, and has two lifts to the fibre cover. A lifted residual component contains precisely two lifted endpoint labels. Its canonically owned edge mass equals the number of original-boundary edges of the underlying path. This number is zero if and only if the two port endpoints represent the same actual corner.*

Proof. A boundary circle containing $q > 0$ distinct ports becomes q compact intervals when their open interiors are removed. Consecutive port edges can meet at one vertex, giving a point interval. If the entire circle consists of one port edge, deleting its interior gives one point with two distinct endpoint occurrences. Summing over active circles gives p residual paths.

An interval bundle restricted to a contractible path is a product, so the corresponding two-sheeted fibre-orientation cover has two path components. Each contains one lift of each endpoint occurrence. Every edge owner belongs to the same residual class as the endpoints of its edge. By Lemma 5.1, no edge contributes twice, and no edge outside the path contributes to it. Lift the ownership program once on each sheet. Each lifted path therefore receives exactly its underlying edge count. A finite cellulated interval has equal endpoint vertices precisely when it is a point. Positive length cannot return to its initial vertex within a genuine degree-two residual boundary containing a port. Passive circles and silent interior classes carry no endpoint labels. □

The last equivalence is where source geometry matters. In an arbitrary quotient graph a positive-length cycle can have the same initial and final vertex. The scalar moment data alone do not exclude that situation.

5.3 The four-field reference codec

The implementation retains the incoming four-field construction as `codec='four'` for differential verification. For $p > 0$ there are $r = 4p$ lifted terminal labels. Port j, local endpoint $e \in \{0, 1\}$, and local fibre sheet $s \in \{0, 1\}$ receive the distinct label

$$i = 4j + 2e + s + 1.$$

Define

$$S_2 = \frac{r(r+1)(2r+1)}{6}, \qquad h = \operatorname{bitlength}(S_2), \qquad \beta = 2^h. \tag{8}$$

The terminal label i contributes

$$1 + i\beta + i^2\beta^2 \tag{9}$$

at its lifted corner occurrence. Every lifted owner edge contributes β^3. Contributions at the same occurrence are added. Both terminal and ownership contributions are supplied as additive constant-weight intervals to the maintained weighted-orbit API.

The global terminal count, global first moment, and global second moment are all strictly less than β. Hence no carry between the three lower fields is possible, even before one uses the degree-two property. For an orbit of scalar weight w successive divisions by β give

$$w = t + \beta u + \beta^2 v + \beta^3 \ell, \qquad 0 \le t, u, v < \beta.$$

If $t > 0$, Lemma 5.2 gives $t = 2$ and labels

$$d = \sqrt{2v - u^2}, \qquad i = \frac{u - d}{2}, \qquad j = \frac{u + d}{2}. \tag{10}$$

The decoder uses an exact integer square root. It checks a positive square discriminant, parity, label range, unique label usage, and coverage of all r labels. A terminal-bearing histogram row must have orbit multiplicity one, because the same globally unique pair of labels cannot belong to two different orbits. A terminal-free row is ignored only after its lower three fields have been checked to be zero.

5.4 A smaller codec using the native degree-two structure

The default `codec='pair'` exploits the authenticated native boundary contract more closely. Once residual degree two is known, no explicit count field is needed to recover a pair. Moreover, the scalar radices can be chosen using the maximum mass of two labels on one path, instead of the sums of all labels in the entire namespace. This improves the power of p in the scalar-magnitude bound from nine to three.

Put

$$A = 2^{\mathrm{bitlength}(2r)}, \qquad s_i = i + Ai^2, \qquad D = 2^{\mathrm{bitlength}(2(r + Ar^2))}. \tag{11}$$

The terminal label i contributes s_i and every lifted owner edge contributes D. Both radices are powers of two, so the implementation uses integer shifts and masks.

Lemma 5.3 (Path-local no-carry decoding)**.** *For a lifted residual path with endpoint labels $i < j$ and edge length ℓ, its scalar orbit weight is*

$$W = \ell D + s_i + s_j.$$

It determines the endpoint pair and length by

$$\ell = \lfloor W/D \rfloor, \quad R = W \bmod D, \quad S = R \bmod A, \quad Q = \lfloor R/A \rfloor, \quad i, j = \frac{S \mp \sqrt{2Q - S^2}}{2}. \tag{12}$$

The remainder is positive on every active lifted path and is zero on every terminal-free orbit.

Proof. The source degree-two theorem supplies exactly two distinct labels on each lifted active path. Equation (11) gives $A > 2r \ge i + j$ and $D > 2(r + Ar^2) \ge s_i + s_j > 0$. Consequently neither radix receives a carry from the fields beneath it: $R = s_i + s_j$, $S = i + j$ and $Q = i^2 + j^2$. The quadratic identity $2Q - S^2 = (j - i)^2$ proves the decoder. Terminal-free orbits consist only of edge mass, which is an integer multiple of D. Nonnegative edge mass and positive terminal labels rule out cancellation. □

This is a path-local radix proof. The sum of terminal weights over the whole namespace need not be smaller than D, and the implementation never assumes that it is. The checker still requires multiplicity one on active histogram rows, exact positive square discriminants, admissible roots,

full label coverage, consistent deck-paired lifts and equal lengths on the two lifts. The source geometry, query weights and orbit proof are independently authenticated before the decoded words are accepted. The smaller codec does not furnish a degree-two geometric-boundary certificate for an arbitrary quotient graph.

There is nevertheless a useful stronger arithmetic safeguard. Completeness of the recovered matching, together with the known input label inventory, replaces the explicit occupancy coordinate even when one does not assume degree two in advance.

Lemma 5.4 (Global inventory forces pair occupancy)**.** *Consider an arbitrary equivalence relation on a finite occurrence namespace. Each of the r distinct positive labels occurs exactly once, with scalar weight s_i from (11); all other weight contributions are nonnegative integer multiples of D. Suppose a complete, exact weighted orbit histogram passes the following arithmetic checks:*

1. *every row of nonzero remainder modulo D has orbit multiplicity one and decodes to two distinct roots in $\{1, \dots, r\}$;*
2. *the decoded pairs are disjoint and exhaust this label set.*

Then every orbit containing labels contains exactly two of them, no label-containing orbit has zero remainder, and every decoded pair is its actual pair of labels. If H is the sum of its nonterminal scalar contributions, its decoded quotient satisfies $\lfloor W/D \rfloor = H/D$.

Proof. An orbit with no labels has zero remainder modulo D. An orbit with only label i has remainder s_i, since $0 < s_i < D$. Since $i < A$, its decoded moment coordinates are $S = i$ and $Q = i^2$, whose quadratic roots are $0, i$. Such a row cannot pass the requirement of two positive roots. Hence every accepted nonzero row represents one actual orbit containing at least two labels.

The exhaustive disjoint decoded inventory contains exactly $r/2$ pairs. Their $r/2$ actual orbits therefore consume at least r actual labels. There are precisely r labels in the whole input, counted once each. Equality follows: every accepted row contains exactly two labels, and no labels remain in a row of zero remainder or in any other orbit. Now the path-local arithmetic inequalities used in Lemma 5.3, which depend only on having two labels, apply to every accepted row. They recover its actual two labels and the sum of the integer coefficients of D in its nonterminal contributions. □

The complete authenticated histogram and the exact initial inventory are essential to this argument. Local moment roots alone are insufficient: the three labels $\{1, 2, 6\}$ have the same first two moments as the pair $\{4, 5\}$. The global argument excludes such a row in a complete accepted partition by counting actual label consumption. This is an arithmetic soundness statement. It does not turn a two-label orbit in a branched graph into a geometric interval, or prove that its arbitrary mass counts genuine boundary edges. Those conclusions still require the independently verified source geometry and canonical ownership theorem.

Theorem 5.5 (Native first-return bridge)**.** *For a valid authenticated native parallelity packet with $p > 0$ ports, one scalar weighted orbit histogram on $\widetilde{X}$ determines:*

1. *the first-return matching of all $2p$ base endpoint occurrences;*
2. *the local fibre reversal and the exact original-boundary edge length of each matched gap;*
3. *the signed cyclic port words, their gap order, and the decomposition of the core attachment into proper intervals and whole boundary circles.*

After forgetting the newly added edge lengths, the output equals the maintained native boundary-word output, with the same canonical port ordering and local chart convention. No separate corner census is needed. For $p = 0$ the cyclic interface is empty and no new census is performed.

Proof. Use either the four-field decoder (10) or the default decoder (12) to obtain the perfect matching on lifted labels. For a base endpoint a, write the mate of its sheet-zero lift as $2b + f$, now

using zero-based lifted labels. The mate of its other lift is $2b + (1 - f)$. These two relations recover the base mate b and fibre flag f. The two lifted path lengths agree, and the implementation checks this equality explicitly. It also checks $b \neq a$ at the level of endpoint occurrences; actual corner equality remains allowed.

By Lemma 5.2, zero length means the two ends share a corner, and positive length means original boundary is present. These are exactly the data recovered by the maintained compiler's separate corner query and lifted endpoint-mask query. To recover cyclic order, cross a port and then its following matched gap. This permutation on oriented endpoint occurrences traverses each active boundary circle in one of its two directions. Start with the least unused port in its positive local chart, matching the maintained convention. Every port appears once. The parity of the gap fibre flags around a complete circle must be zero; this is checked rather than inferred from rank slopes.

If every gap in a word has length zero, the attachment is the whole circle. Otherwise, begin immediately after every positive-length gap and collect successive ports across zero-length gaps. These maximal runs are exactly the proper interval attachments. Their endpoint tokens and fibre parity are read from the same matching and ordered gap flags. Thus all fields of the existing skeleton are identical, while the new positive integer lengths refine its original-boundary-present booleans. Neither traversal requires reconstructing endpoint masks. The case $p = 0$ is immediate. □

What the local charts do not provide. The chosen direction on each word is a canonical reporting convention. It is not a coherent boundary orientation of all the circles belonging to one orientable base. The native packet supplies local port charts and fibre transitions; a later surgery algorithm requiring coherent base frames must derive or receive those frames separately. Similarly, the word does not retain an arbitrary additional boundary marking lying inside a positive-length gap.

5.5 Encoded cost and the actual implementation claim

Proposition 5.6 (Input scalar and output-size bounds)**.** *For the default pair codec with $p > 0$, every additive input weight has $O(\log(p+1))$ bits. Its total scalar mass on the double cover is less than*

$$2(E+p)D \leq 2ND = O(N(p+1)^3). \tag{13}$$

Every active orbit separately has scalar weight less than $(N+1)D$. Consequently every orbit weight uses $O(\log(N+1) + \log(p+1))$ bits, with only three logarithmic powers of p in the scalar bound. The gap and word portion of the returned cyclic interface uses

$$O(p(\log(N+1) + \log(p+1)))$$

bits. The complete skeleton also carries source core-handle identifiers; its bound is $O(p(\log(N+1) + \log(T+1) + \log(p+1)))$, excluding discovery certificates and the optional base-profile ledger. Its assembly uses $O(p)$ endpoint and gap records and never materializes an $O(p)$-bit incidence mask.

Proof. Equation (11) gives $2r < A \leq 4r$ and

$$2(r + Ar^2) < D \leq 4(r + Ar^2) \leq 20r^3.$$

Every terminal contribution satisfies $s_i < D/2$. There are $r = 4p$ terminal contributions and $2E$ lifted owner edges. The total scalar mass is therefore less than

$$2ED + (r/2)D = 2(E+p)D \leq 2ND.$$

An active orbit has at most E owner edges and terminal remainder less than D, giving the stated per-orbit bound. These inequalities also bound the input scalar bit lengths.

There are p gaps, each with two endpoint identifiers, one fibre bit and one length at most E. The cyclic words contain p port occurrences in total. The attachment words also contain each port once, so their total length is p. Source port ranks have $O(\log(N+1))$ bits, while the source core-handle identifiers have $O(\log(T+1))$ bits. These facts give the two output bounds. The latter term cannot be discarded merely because only a few tetrahedra contain nonzero normal coordinates. The direct matching and successor traversals need only a bounded number of arrays or dictionaries indexed by the $O(p)$ endpoints and gaps. □

For comparison, the retained four-field codec has $\beta \le 2r^3$ and total scalar mass less than $(2E+1)\beta^3 = O((N+1)(p+1)^9)$. Both encodings have the same big-O binary length, but the path-local codec reduces the number of label bits and scalar-arithmetic constants. The complete decoded outputs agree.

If there are g source gluing records and a original-boundary free-edge records, the double-cover query has $4g+2a$ pairing records and $4p+2a$ additive weight records. A simple endpoint sweep initially canonicalizes these additions into $O(a+p)$ weight runs. The source construction makes $g+a+p = O(T)$, with binary endpoints. Exact length extraction therefore preserves the polynomial encoded size of the source query.

There is a practical tradeoff. The maintained sparse-mask compiler has input weights only at marked endpoints; its previously established uniform sparse-token run bound does not apply to the new edge-mass query. Original-boundary owner intervals add weight runs. The initial bound is $O(a+p)$, not a port-only bound, and the measured maximum number of runs can be larger even though one entire census has been removed.

The weighted Agol–Hass–Thurston theorem supplies a polynomial algorithm for the encoded weighted interval-orbit task [1, Theorem 16]. Composing that theorem with the explicit reduction above gives a polynomial bound for this native first-return problem in the source record count and binary lengths. This is an algorithmic existence and reduction statement. We do not infer a new uniform polynomial bound on every intermediate weight run or every certificate trace of the particular maintained Python implementation. For that implementation the package supplies independently replayed executions, cancellation contracts, and concrete measurements. Those executions are evidence about the implementation; they are not a proof of its uniform worst-case trace complexity.

5.6 Certificates, owner profiles, and failure contracts

The direct API is `normal_parallelity_moment_words`. Its certificate binds the selected codec and contains the source fingerprint, the complete cyclic skeleton and one weighted census certificate. The checker reconstructs the geometry through the maintained raw-cut reference path, independently builds the lifted affine pairings and ownership weights, replays the weighted trace, decodes the moment pairs by a separate implementation, and reconstructs the words using a separate permutation traversal. It imports neither the new producer nor its query builder, decoder or assembly routine. Tests disable those functions, both native discovery schedulers, and the production geometry builder while the checker replays a serialized binary certificate.

The optional `moments_from_parallelity_packet` API accepts an existing full normal-cut parallelity certificate. It first verifies that packet, including its four inherited discovery traces, then compiles one new moment census from its authenticated free-edge inventory. Source authentication is additional work; it is not hidden in the one-new-census claim. Its combined certificate retains the upstream proof and binds every active loop to its certified base-profile owner. For a profile with boundary count b and k active loops, the passive count is $b-k$. Port-bearing profiles have multiplicity one; port-free profiles retain their binary multiplicity and passive count per copy. In

this way even passive components absent from the terminal census are preserved.

Discovery exhaustion returns `INCONCLUSIVE` without a skeleton or certificate. The direct replay cap counts its one trace; the packet replay cap counts the sum of the four inherited traces and the new trace. Malformed proofs return false, whereas cooperative cancellation and invalid-source exceptions propagate. These statuses are not knot verdicts. The new proof authenticates a compressed boundary operation, not the validity or completeness of a proposed recognition hierarchy.

5.7 Native verification and paired measurements

The verification passed eight native test groups in 113.590 seconds and one additional arithmetic test group in 0.026 seconds. Complete cyclic outputs were compared on all 157 inherited source cases among the existing native compiler and both new codecs. Each new certificate was independently replayed. The edge-by-edge oracle could expand 146 of these cases; it checked 219 positive-length gaps and 3420 point gaps. The bank includes five sources whose parallelity polygons are exclusively edge caps, 181 whole-circle attachment components, and 546 gaps with nontrivial local fibre transport. Additional cases cover layered-source multiples, one-sided bases, source-packet authentication and JSON transport, altered moments and lengths, source mismatch, exact resource caps and cancellation.

The arithmetic test enumerates all 4140 set partitions of eight terminal labels, adds arbitrary nonnegative edge masses and silent repeated orbits, and applies the no-count decoder. It accepts exactly the 105 perfect matchings and recovers their true pairs and masses. All partitions containing a singleton or an overfull terminal class are rejected. This is a finite regression check of Lemma 5.4, independent of the geometric source-bank coverage.

For the $T = 2$ layered source scaled by $m = 2^b$, with $b = 32, 512, 16384$, the new census retained two ports, 60 lifted ownership intervals, 68 additive weight records and 131 orbit cycles in every case. This checks binary-coordinate handling without expansion. The active lengths in this family are both seven; the test is not presented as a family of growing active gap lengths. The unscaled $T = 16$ source does have a gap of 39787 edges, recovered without following them individually.

Table 4 reports five measurements per implementation after one untimed warmup per implementation and source. The execution order alternates existing–pair–four and four–pair–existing. Every timed discovery call starts with the original triangulation and coordinates and includes source validation, geometry construction, both or one orbit census respectively, complete cyclic assembly and certificate construction. No source geometry or orbit result is reused. The independent source-bound replay is timed separately. These are local measurements on Python 3.12.14 under Linux x86-64 in the shared execution environment, not hardware-isolated estimates or whole-knot recognition benchmarks.

		Discovery, ms			Replay, ms		
Case	p	Existing	Four	Pair	Existing	Four	Pair
Layered $T = 2$	2	2.805	3.016	2.968	2.900	3.254	3.033
Layered $T = 4$	15	69.697	67.925	70.650	50.308	63.959	62.811
Layered $T = 8$	47	589.207	531.740	521.412	418.031	406.742	439.730
Layered $T = 16$	111	3754.319	3034.441	3057.760	2505.085	2278.534	2212.595
$T = 2$, $m = 2^{32}$	2	40.726	36.990	33.990	20.270	28.035	29.060
$T = 2$, $m = 2^{512}$	2	34.477	33.111	33.310	16.373	25.984	26.983
$T = 2$, $m = 2^{16384}$	2	135.094	118.898	117.586	64.402	89.302	85.802
Cap-only source	8	0.454	0.434	0.461	2.007	1.726	1.686
Port-free binary	0	0.318	0.276	0.280	0.976	0.980	0.894

Table 4: Median complete source-query and independent replay times. Four denotes the reference four-field codec and Pair the default path-local codec. Both new codecs also certify exact gap lengths.

Case	Orbit cycles	Peak runs	Scalar bits	Certificate KiB
Layered $T = 2$	$42 \to 37$	$13 \to 21$	$25 \to 14$	$5.8 \to 5.2$
Layered $T = 4$	$250 \to 189$	$77 \to 161$	$52 \to 21$	$45.9 \to 37.3$
Layered $T = 8$	$690 \to 485$	$265 \to 357$	$67 \to 27$	$160.7 \to 113.7$
Layered $T = 16$	$1570 \to 1077$	$649 \to 741$	$76 \to 30$	$436.3 \to 280.3$
$T = 2$, $m = 2^{32}$	$168 \to 131$	$17 \to 85$	$25 \to 14$	$37.5 \to 31.8$
$T = 2$, $m = 2^{512}$	$168 \to 131$	$17 \to 85$	$25 \to 14$	$192.9 \to 164.5$
$T = 2$, $m = 2^{16384}$	$168 \to 131$	$17 \to 85$	$25 \to 14$	$4435.1 \to 3782.4$
Cap-only source	$2 \to 1$	$16 \to 16$	$39 \to 18$	$3.1 \to 2.7$
Port-free binary	$0 \to 0$	$0 \to 0$	–	$0.3 \to 0.3$

Table 5: Existing-to-pair structural measurements, except Scalar bits, which compares the maximum additive input-weight bit length for four-field versus pair codecs. Orbit cycles sum the two existing censuses; Peak runs takes their maximum. Certificate size is compact JSON, using hexadecimal transport for sufficiently large integers. These are observed sizes, not uniform worst-case bounds.

Compared with the existing native compiler, pair-codec discovery is about 1.13 times faster for $T = 8$ and 1.23 times faster for $T = 16$ in this run. The $T = 2$, $T = 4$ and cap-only examples are slightly slower. Replay is not uniformly faster: on the $m = 2^{512}$ example it increases from 16.373 to 26.983 ms, whereas on the $T = 16$ example it decreases from 2505.085 to 2212.595 ms. The new ownership weights increase the maximum run count, especially in the binary two-port family, from 17 to 85.

The pair codec reduces the maximum additive scalar from 76 to 30 bits on the 111-port example and from 25 to 14 bits on the two-port binary family. Its wall-clock differences relative to the four-field codec are small and mixed, as the table records. Its principal justification is the proved cubic scalar bound and the removal of the unnecessary count field; a further uniform practical speedup is not claimed. The complete single samples, proof sizes and representative replayable certificates are included in the results directory.

The native source bank contains shared-corner point gaps and whole-circle attachments, but none of its explicitly expanded examples has a one-edge boundary loop. A separate geometric regression therefore tests a triangle whose first two edges are identified in reverse, leaving its third edge as a one-edge boundary circle. Marking that edge must give one zero-length gap and one circle attachment. That test exercises the actual native moment query and weighted proof replay, compared with a literal incidence walk. It is explicitly a geometric packet regression, not an additional authenticated source triangulation.

The files `native_moments.py` and `native_moments_verify.py` contain the producer and checker; `native_moments_literal.py` supplies the separate small-input edge walker. The source bank records its upstream commit, source path and SHA-256 digest, and omits unrelated upstream proof payloads. Representative certificates cover a nontrivial layered source, a cap-only source, and a source with multiplicity $2^{16\,384}$. They can be replayed without rerunning discovery.

Resulting research obligation. The missing native ownership-to-moment bridge from the incoming report is now concrete and source authenticated. The next performance question is whether exact edge mass admits an instance-independent run bound strong enough to prove a uniform resource bound for this particular replay implementation. The next geometric question is how to supply coherent base frames and any additional pattern data needed by subsequent surgery. Neither question is resolved by agreement of the present cyclic outputs.

6 Reconstructing an unmarked cut manifold

Large normal coordinates do not by themselves force a large triangulation of the manifold obtained by cutting. A long stack of parallel normal discs is represented by an interval bundle, whose topological complexity may be much smaller than the number of discs in the stack. There are nevertheless two distinct costs: the topology of the interval-bundle bases, and the information in the attaching maps. The purpose of this section is to state an explicit reconstruction bound that accounts for both.

The result below is a constructive consequence of surface classification and the published parallelity-bundle machinery. We do not claim it as a new major topological theorem. Its contribution here is to make the output convention, the cellular input requirement, and the limits of its use in a hierarchy algorithm explicit.

For any compact surface F, orientable or otherwise, set

$$\chi_-(F) = \sum_{F_0 \in \pi_0(F)} \max\{0, -\chi(F_0)\}.$$

The word *unmarked* means that the output is required to have the correct homeomorphism type, without a prescribed parametrisation of its boundary or a prescribed identification with a subspace of the original manifold.

6.1 A reconstruction bound with a supplied cellular core

Theorem 6.1 (Reconstruction from explicit core and bundle data)**.** *Suppose that a compact orientable 3-manifold, with corners allowed, is specified as*

$$X = C \cup \bigcup_{j=1}^{m} B_j, \qquad B_j \longrightarrow F_j,$$

where the F_j are compact connected surfaces and the B_j are pairwise disjoint interval bundles with orientable total spaces. Assume the following input data.

1. *The core C has an explicit cellular decomposition with total size at most q, using a fixed finite collection of 3-ball templates. The incidence data and the cellular attaching maps are included in this size. Equivalently, one may supply an explicit triangulation of C with $O(q)$ tetrahedra, together with its boundary subdivisions.*
2. *Each base F_j is specified by its orientability, genus or crosscap number, and number of boundary components. The interval-bundle type is specified. There are no identifications between horizontal boundaries of the B_j in this presentation.*

3. *Each $B_j \cap C$ is a subset of the vertical boundary $\partial_v B_j$. Its attaching data consist of explicitly ordered fibre rectangles and their cellular identifications with rectangles on ∂C. Boundary gaps, horizontal-sheet incidence, and the relevant orientations are recorded. The total size of this data and of the required subdivisions of ∂C is included in q.*

The presentation is assumed to be valid; recognising validity of arbitrary cellular presentations is not part of the assertion. Then X admits an ordinary, unmarked triangulation with

$$O\left(q + \sum_{j=1}^{m} (1 + \chi_-(F_j))\right) \tag{6.1.1}$$

tetrahedra. Such a triangulation can be constructed in time polynomial in the input bit length and the displayed bound.

Every component and every repeated copy is counted in the sum. If a family of detached, identical bundles is instead specified by a binary multiplicity, expanding that family incurs the corresponding output cost.

Proof. First refine the fixed core templates and their attaching rectangles into triangles and tetrahedra. A fixed template requires a fixed number of tetrahedra; the stipulated cellular size includes all necessary boundary subdivisions. Consequently this stage has cost $O(q)$.

We use the following elementary surface-construction fact. A connected compact surface F, together with v prescribed subdivision vertices on its boundary, has a triangulation respecting those vertices with

$$O(1 + \chi_-(F) + v)$$

triangles. To see this, choose the usual polygon model from the classification of compact surfaces, triangulate it, and subdivide the boundary as prescribed. For an orientable surface with b boundary components, $\chi(F) = 2 - 2g - b$; in the nonorientable case, $\chi(F) = 2 - k - b$, where k is the crosscap number. These formulas bound the size of a polygon model by $O(1 + \chi_-(F))$, apart from the prescribed boundary subdivisions. The sphere, disc, annulus, projective plane, and Klein bottle are covered by fixed templates. A bounded subdivision converts any polygon identifications into a genuine surface triangulation. Prescribed cyclic orders on boundary components are respected throughout.

Apply this construction to each F_j, introducing the subdivision points dictated by the ordered attaching rectangles. Their total number is $O(q)$. Boundary components with no attaching rectangles require only a bounded number of additional vertices each; their number is already bounded by $O(1 + \chi_-(F_j))$. Thus the total number of base triangles is

$$O\left(q + \sum_{j} (1 + \chi_-(F_j))\right).$$

An interval bundle over a triangle is a triangular prism. Glue these prisms using the specified interval-bundle type. For an orientable total space, the twisting character of the bundle is the orientation character of its base: the bundle is a product over an orientable base, and is the orientable twisted interval bundle over a nonorientable base. The polygon construction therefore supplies the needed transition maps directly. Subdivide prism faces compatibly and triangulate every prism with a bounded number of tetrahedra. For example, first barycentrically subdivide the polygonal faces and then cone each subdivided prism boundary to an interior vertex; this avoids choices of incompatible diagonals. The attaching rectangles match the refined core cells after a bounded further subdivision. Gluing gives the required triangulation of X, with the asserted bound.

All constructions are finite operations on explicit cells, together with integer arithmetic on the supplied topological parameters. Writing down the polygon models and the tetrahedra takes time

polynomial in the input bit length plus the output size. Binary multiplicities can be retained in a compressed representation; if the requested output is an ordinary triangulation, the corresponding copies must be written out. □

Why counting handles alone is insufficient. The hypothesis is a bound on *cellular size*, including attaching maps. It is not a bound on the number of abstract handles whose attaching words might have unbounded length. An arbitrary gluing map given by a short word in large powers of mapping classes does not satisfy the hypothesis merely because its word has few syllables. The theorem applies to the explicit ordered fibre rectangles in its statement.

There is no additional unrecorded torus-slope parameter in this particular presentation: the bundles attach through their vertical annuli and rectangles, while their horizontal boundaries remain unglued. For example, a Dehn twist on a vertical annulus that fixes its boundary extends over a collar of that annulus in the bundle. In collar coordinates $S^1 \times [0, \varepsilon] \times [0, 1]$, with the vertical annulus at $r = 0$, an extension of k twists is

$$(\theta, r, s) \longmapsto (\theta + 2\pi k s f(r), r, s), \qquad f(0) = 1, \quad f(\varepsilon) = 0.$$

It changes the parametrisation on a horizontal boundary collar. This is permitted for an unmarked output, and explains exactly why the same argument does not provide a reconstruction relative to fixed horizontal seam charts.

6.2 Controlling the bases by the cutting surface

The following elementary estimate converts the topological size of the bundle bases into a parameter of the cutting surface.

Lemma 6.2 (Subsurface complexity)**.** *Let Σ be a compact surface and let $A \subset \Sigma$ be a compact subsurface of dimension two. After rounding corners, suppose the relative frontier of A in Σ is a disjoint union of c circles and properly embedded arcs. Write*

$$\kappa(Y) = b_0(Y) + b_1(Y; \mathbb{Q})$$

for any compact surface Y. Then

$$b_0(A) \le b_0(\Sigma) + c, \qquad b_1(A; \mathbb{Q}) \le b_1(\Sigma; \mathbb{Q}) + c,$$

and consequently

$$\kappa(A) \le \kappa(\Sigma) + 2c.$$

If Σ is orientable, then also

$$\kappa(\Sigma) \le \chi_-(\Sigma) + 3b_0(\Sigma).$$

Proof. Cutting a surface along one properly embedded arc or circle increases the number of components by at most one. The closures of the components on the A side form a subcollection of the components obtained by cutting along the frontier. This proves the bound for $b_0(A)$.

Let D be the closure of the complement of A. Use small collar neighbourhoods to obtain a Mayer–Vietoris decomposition of Σ whose intersection retracts onto the frontier. Its first rational Betti number is at most c. Exactness of

$$H_1(A \cap D; \mathbb{Q}) \longrightarrow H_1(A; \mathbb{Q}) \oplus H_1(D; \mathbb{Q}) \longrightarrow H_1(\Sigma; \mathbb{Q})$$

shows that $b_1(A; \mathbb{Q}) + b_1(D; \mathbb{Q}) \le b_1(\Sigma; \mathbb{Q}) + c$. Discard the nonnegative term $b_1(D; \mathbb{Q})$ and add the component bound. Finally, the classification formulas for a connected orientable surface, including the closed and nonclosed cases, give $1 + b_1 \le \chi_- + 3$. Summation over components completes the proof. □

Corollary 6.3 (Size of an unmarked normal cut)**.** *Let M be a compact orientable 3-manifold given by a triangulation with t tetrahedra. Let S be a compact orientable normal surface of weight W, specified by its binary normal coordinates. Retain all components of*

$$M_S = M \setminus \operatorname{int} N(S).$$

Then an ordinary unmarked triangulation of M_S with

$$O(t + b_0(S) + \chi_-(S))$$

tetrahedra can be constructed in time polynomial in

$$t, \quad \log(2 + W), \quad b_0(S) + \chi_-(S).$$

No irreducibility or essentialness assumption is needed for this cutting statement. The running-time assertion includes the cost of expanding all output components.

Proof. Use the parallelity bundle $\mathcal{B}$ for the pair consisting of M_S and the two copies $S^+ \sqcup S^-$ of S in its boundary. The compressed-cut algorithm of Lackenby gives the remaining core, the base-surface types, and ordered vertical attachment data in time polynomial in $t \log(2 + W)$; see Theorem 9.3 and the explanatory paragraphs and proof in Section 9.4 of [2]. In a tetrahedral source, the nonparallel pieces and attachment squares have bounded complexity in each local tetrahedral cell. Summing over these cells gives $O(t)$ total cellular incidences, including the subdivisions of vertical attachments. An individual dual two-handle may have many incidences when its original edge has high valence; only the total incidence bound is asserted here. Accordingly, the construction supplies the explicit core and attachment complexity $q = O(t)$ needed in Theorem 6.1; no bound on handle count alone is being substituted. Repeated detached components are retained with multiplicities until expansion is requested.

Set $A = \partial_h \mathcal{B} \subset S^+ \sqcup S^-$ and let F_j be the connected base surfaces. The relative frontier of A in $S^+ \sqcup S^-$ is incident to the explicit nonparallel core. Each of its arcs or circles contains an edge of the vertical attachment data, and there are only $O(t)$ such edges. Portions lying in the original boundary of $S^+ \sqcup S^-$ are not part of this relative frontier. Thus its number c of components is $O(t)$. Components of $\mathcal{B}$ with no core attachment are allowed in this argument: they contribute whole components to A, rather than additional frontier.

Apply Lemma 6.2 to obtain

$$\kappa(A) \le 2\kappa(S) + O(t) = O\big(t + b_0(S) + \chi_-(S)\big).$$

The horizontal boundary A_j of the bundle over F_j is its two-sheeted boundary cover. Hence

$$\chi_-(A_j) = 2\chi_-(F_j), \qquad b_0(A_j) \ge 1.$$

For every compact surface, χ_- is at most its first rational Betti number. Consequently

$$\sum_j (1 + \chi_-(F_j)) \le b_0(A) + \tfrac{1}{2}\chi_-(A) \le \kappa(A).$$

Theorem 6.1 now gives the tetrahedron bound and the construction time. The weighted orbit machinery can keep identical component data grouped with binary multiplicities throughout the compressed part of the computation; writing out their copies costs only the explicit output parameter in the statement. □

A necessary component term. The term $b_0(S)$ cannot simply be dropped from a theorem retaining the whole cut manifold. Fix a triangulation of a 3-ball having an interior vertex, and let v be the normal vector of its vertex-linking sphere. The vector Nv represents N nested parallel normal spheres and has bit length $O(\log(2+N))$. Cutting along them produces $N+1$ nonempty 3-manifold components. An ordinary triangulation of their disjoint union contains at least $N+1$ tetrahedra, although every cutting component has $\chi_- = 0$. One may retain these product regions by a compressed multiplicity, or expand them and pay their output size; these are different output specifications.

What this does and does not supply to a hierarchy. For a connected disc, annulus, sphere, or torus, the displayed unmarked size bound is $O(t)$. More generally, it replaces dependence on raw normal weight by dependence on the topological size of the cutting surface. This is useful when a prospective hierarchy supplies a bound on $b_0(S) + \chi_-(S)$ even though the normal coordinates are very large. Repeatedly applying the corollary, however, does not by itself give a hierarchy algorithm of quasipolynomial complexity. It supplies neither a bound on the number of cuts nor a controlled identification of their horizontal boundaries with previously stored surfaces. In particular, transferring an existing boundary pattern, preserving normal-coordinate charts for later cuts, and retaining the pairing that reglues S^+ to S^- require additional information. A future implementation should therefore expose two separate interfaces: unmarked reconstruction with the bound proved here, and reconstruction relative to explicitly costed boundary markings. The latter remains a research obligation.

7 What a quasipolynomial completion must prove

The preceding operations address different terms in a hierarchy's cost. The genus budget limits the network joining retained pieces. The native moment compiler limits the description of a boundary response. Unmarked reconstruction limits the size of one new triangulation when the cutting surface has controlled topological complexity. None of these statements bounds the number of hierarchy states visited, or the cost of discovering the next useful surface. This section makes that remaining distinction quantitative.

7.1 Four costs that must be bounded together

For a state H, write $s(H)$ for its complete encoded size, including its manifold presentation, boundary pattern, normal-coordinate charts, compressed component multiplicities, and any proof data the algorithm must retain. This is deliberately more than the tetrahedron count. A prospective algorithm must bound the following quantities.

1. *Search volume:* the number $J(n)$ of state visits or elementary hierarchy iterations processed on an n-crossing input. A repeated visit is counted again.
2. *State volume:* the maximum value $S(n) = \max_H s(H)$ during that computation, including the largest binary integer lengths.
3. *Local work:* the work needed to discover, perform and verify the permitted transition from a state of size s. A polynomial-time checker for a supplied surface does not bound its discovery cost.
4. *Completion:* a proof that every permitted branch is handled, that the selected hierarchy operations remain sound, and that the final test distinguishes all knots. A resource cap returning `INCONCLUSIVE` is an honest software contract, but not completeness.

If local work is at most $S(n)^a$ for a fixed constant a, then the direct accounting bound is

$J(n)S(n)^a$, plus input construction. This elementary product is useful because it exposes exactly where a claimed fast local primitive can and cannot change the overall exponent.

Theorem 7.1 (Conditional hierarchy accounting)**.** *Suppose a complete and correct recognition procedure on diagrams with $n \geq 2$ crossings has parameters $L = L(n) \geq 1$ and $G = G(n) \geq 1$. Assume its input construction takes polynomial time in n, and that all subsequent work, including its final decision, is charged to the iterations below. Assume fixed constants $a, c, C > 0$ and $A > 1$ satisfy all of the following.*

(i) It performs at most $C(L+1)(G+1)^L$ elementary iterations, counting revisits to a state again.

(ii) $G \leq n^c$, and every encoded state and transition output has size at most $n^c A^{cL}(L+1)^c$.

(iii) Discovery, source validation, exact local computation, certificate production and replay for one iteration together cost at most the ath power of this size, times a fixed constant.

Then its total running time is

$$\exp\big(O((L+1)\log n + L + \log(L+1))\big).$$

In particular, $L = O((\log n)^2)$ gives $2^{O((\log n)^3)}$, while $L = O(\log n)$ gives $n^{O(\log n)}$.

Proof. Multiply the bounds in (i) and (iii), using (ii), and take logarithms. Up to constant factors, the logarithm of the iteration cost is at most

$$\log(L+1) + L\log(G+1) + ac\log n + acL\log A + ac\log(L+1).$$

Since $\log(G+1) = O(\log n)$ and A, a, c are fixed, this is the stated expression. Adding the polynomial input-construction cost does not change it, since that cost has logarithm $O(\log n)$. Substituting either bound on L proves the last two claims. □

This is an implication, not a new hierarchy-existence theorem. Its first assumption matches the form of the iteration bound in Proposition 9.1 of the 2026 preprint [7]; a finite search tree of depth L and branching at most $G+1$ also has the stated coarse bound. The geometric meanings of L and G must come from the actual procedure, not from arbitrary tuning parameters. The theorem explains why the two 2021 depth announcements lead to genuinely different exponents.

7.2 What the unmarked reconstruction bound contributes

Proposition 7.2 (Iteration of an unmarked size bound)**.** *Suppose a sequence of unmarked reconstructions satisfies*

$$t_{i+1} \leq A(t_i + k_i + 1), \qquad A \geq 1,$$

where $k_i = b_0(S_i) + \chi_-(S_i)$ counts every cutting component. Then

$$t_i \leq A^i \left(t_0 + \sum_{j<i} (k_j + 1) \right).$$

If $t_0, k_j \leq n^c$ and $i \leq L$, then

$$\log(t_i + 1) = O(L + \log(L+1) + \log n)$$

for fixed A, c.

Proof. For $i = 0$ the assertion is equality. Given the bound for i, multiply by A and use $A \leq A^{i+1}$ to absorb the next additive term. This proves the first inequality by induction. The sum is at most $i(n^c + 1)$, which gives the logarithmic estimate. □

Thus even a constant-factor expansion at each of $O((\log n)^2)$ cuts can remain compatible with the later announced quasipolynomial exponent. A perfectly size-preserving reset is not necessary for this accounting. What is necessary is control of the *whole* state: the statement above does not bound the word describing a carried boundary pattern, the normal coordinates of the next surface, or the search required to find it. If the remainder of each state has *total* encoded size polynomial in t_i and n, the encoded-state part of assumption (ii) in Theorem 7.1 follows after adjusting the fixed constants. A polynomial bound for G and the discovery-time bound in (iii) remain separate obligations. Small output size alone supplies neither of them.

Nor can one discard multiplicities to make k_i appear small. The nested-sphere example in Section 6 shows that a single type record can represent exponentially many components. A hierarchy may have a valid rule for purging those components; otherwise their explicit reconstruction is a real output cost. Any proposed reset must state which of these two contracts it implements.

7.3 Sharing already supplied transport along a forest

The rooted forest of Section 3 gives a precise place to store transport information without expanding every source route. There is an elementary algebraic principle behind this design.

Proposition 7.3 (Forest transport circuit)**.** *Let Q be a finite oriented graph with a rooted spanning forest F, and suppose every oriented edge e has a supplied label a_e in a group, with $a_{\bar{e}} = a_e^{-1}$. Labels may instead be words in a formal group alphabet, represented by shared expression nodes. There is a circuit with $O(|V(Q)| + |E(Q)|)$ additional product and inverse gates that labels every nonforest edge of the contracted graph and preserves the products around closed walks based at roots. Only the original label descriptions are needed as input leaves.*

If group multiplication and inversion have unit cost, all contracted labels can be evaluated using $O(|V(Q)| + |E(Q)|)$ group operations. The claim is about supplied algebraic transport; it does not construct a geometric transport map on a boundary pattern.

Proof. Let P_v be the product along the forest route from v to its root, with products written in path order. For a root set $P_v = 1$; otherwise define a shared gate

$$P_v = a_{(v,\mathrm{parent}(v))}\, P_{\mathrm{parent}(v)}.$$

For an edge $e : u \longrightarrow v$ not in the forest, assign

$$a'_e = P_u^{-1} a_e P_v.$$

Each vertex requires a bounded number of gates and each retained edge requires a bounded number more. A forest edge has transformed label one. Along any walk, the consecutive $P_v P_v^{-1}$ factors cancel, leaving P_{start}^{-1}, the original path product, and P_{end}. For a closed walk based at a root both outside factors are one. Processing the forest from roots to leaves and then processing the nonforest edges gives the operation count. No path is expanded as a word. □

This supplies a useful interface for a future patterned reducer. Local fibre signs can be evaluated in C_2 immediately. Longer curve or chart instructions can remain as a circuit until a consumer requests them. The missing step is geometric: prove that the proposed source labels encode the required maps, provide compatible domains, and bound the cost of the subsequent equality or isotopy problem. The delivered graph module stores the parent forest; it does not implement arbitrary group evaluation or a mapping-class-group engine. Keeping that distinction prevents an algebraic compression from being mistaken for a solved three-dimensional marking problem.

7.4 Why simultaneous disjoint homology bases are insufficient

The word "multisurface" in the announced architecture cannot be replaced by "a disjoint collection forming a full relative homology basis." The obstruction is already visible in a product.

Proposition 7.4 (A disjointness obstruction)**.** *Let $M = \Sigma_h \times I$, where Σ_h is a closed connected orientable surface of genus $h \geq 1$. Suppose pairwise disjoint, compact, properly embedded, oriented surfaces $S_1, \ldots, S_k$ represent linearly independent classes in $H_2(M, \partial M; \mathbb{Q})$. Then $k \leq h$, although that relative homology group has dimension $2h$.*

Proof. Poincaré–Lefschetz duality identifies their classes with linearly independent classes $\alpha_i \in H^1(M; \mathbb{Q})$. Disjoint representatives have zero mutual intersection, so $\alpha_i \smile \alpha_j = 0$ for $i \neq j$. For $i = j$, the same vanishing follows from graded commutativity over $\mathbb{Q}$. Pullback along the inclusion of $\Sigma_h \times \{1/2\}$ identifies $H^1(M; \mathbb{Q})$ with its $2h$-dimensional surface cohomology and preserves cup products. The cup-product pairing, evaluated on $[\Sigma_h]$, is nondegenerate and alternating. An isotropic subspace of a $2h$-dimensional symplectic space has dimension at most h: its inclusion in its symplectic orthogonal gives $k \leq 2h - k$. This proves the result. □

The product in this example also embeds in S^3, as a small collar of an embedded closed genus-h surface. Thus restricting attention to compact submanifolds of S^3 does not remove the obstruction. This classical obstruction identifies a concrete design requirement. An implementation must support the intersections permitted by the multisurface method, or justify a different hierarchy with equivalent quantitative control. Fast software for a disjoint special case does not settle that requirement.

7.5 The remaining logical gap

The initial torus reducer starts with empty pattern and a verified irreducible knot exterior. A later cut normally introduces new boundary data. Erasing it when absorbing a ball may change the problem being solved, even though the underlying unmarked manifold is unchanged. Likewise, a cyclic word and a fibre flip do not automatically specify a compatible system of boundary charts and all transition maps needed for a gluing operation. A nonorientable base with orientable total interval bundle has no global fibre orientation; local frames and their transition data are the appropriate objects. The output contract of the native compiler deliberately stops before these gluing assertions.

To close the global gap one must prove a complete surface-discovery procedure with the required hierarchy length, establish bounds on the pattern and its transport, and maintain those bounds through every reconstruction and simplification. The current contributions reduce specific costs and remove specific missing source interfaces. They provide no shortcut around these remaining theorems.

8 Further research questions and proposed approaches

The research priorities below are chosen to connect the proved local results to a complete hierarchy. Each has a concrete success criterion. Several are mathematical questions rather than requests for more benchmarking; a fast implementation cannot substitute for their missing proofs.

8.1 Priority I: preserve geometry while shrinking the state

Research problem 8.1 (Patterned ball absorption with shared source addresses)**.** Given a verified unused-cocore decomposition of an irreducible manifold, a boundary pattern in explicit or compressed source charts, and the rooted forest of Theorem 3.4, construct a pattern-preserving

reduced presentation and a verifiable transport map. Bound its encoded size in terms of the source size, the retained genus budget, and a precisely defined pattern-interface parameter.

The promising approach is to retain an absorbed ball only when its pattern has information that cannot be carried through its parent collar by the chosen chart representation. The other balls become shared transport nodes. Proposition 7.3 prevents the forest paths themselves from causing quadratic duplication, once the local transport maps have been constructed. A successful proof must also control the complexity of those maps. Merely storing one Boolean "marked" flag per ball does not encode an embedded graph on its boundary, and a fixed genus does not bound the normal weight of a curve on that boundary.

A useful first target is a pattern that is a disjoint family of arcs, with every endpoint on an explicitly listed attaching disc. The next target allows trivalent vertices and closed components. For each class, one should either prove a polynomial compressed transport bound or exhibit a family forcing the proposed representation to grow. The theorem should state which isotopies are allowed at the retained core and which boundary components remain individually labelled. These are part of the input contract, not implementation details that can be chosen after a proof.

Research problem 8.2 (A normal-coordinate adapter for the retained core)**.** Turn the initial admissible handle subpresentation into a normal or regulated-surface coordinate system whose solutions have an explicit source interpretation. Construct a polynomial-time map for the surfaces and boundary data actually needed by the next hierarchy step.

The four local characteristic graphs give a finite starting catalogue. For each graph, enumerate the allowed local disc types, their boundary arcs, and matching equations. Prove that assembling these records gives exactly the intended embedded surfaces, including the restrictions needed to avoid incompatible local choices. The main challenge is the relation between this inherited handle structure and any new triangulation: a small count of local types is not, by itself, a bound on every global coordinate or a surface-search algorithm. An adapter tested on the canonical trefoil and figure-eight cores would be a concrete first implementation milestone, but its completeness needs a separate proof.

Research problem 8.3 (Reconstruction relative to a specified pattern)**.** Strengthen Corollary 6.3 to an output that retains a specified boundary pattern, seam identification and normal-coordinate charts. State and prove the additional parameter needed beyond $t + b_0(S) + \chi_-(S)$.

There are two useful output models to compare. One emits an ordinary triangulation with the marking expanded into its individual normal arcs; the other retains a small triangulation and a compressed marking or transport circuit. A binary normal-coordinate vector is itself a compressed representation and must be distinguished from the first model. The research should begin by finding families that separate precisely specified encodings, for example repeated twists in a boundary collar. Large normal weight alone does not imply large binary coordinate length. The extension of a vertical-annulus twist used in the unmarked proof explains why the unmarked complexity can stay small while a prescribed marking changes. Algorithms for curves in surfaces [6] are candidates for comparing or normalizing the resulting compressed markings, once the three-dimensional chart transfer has been supplied.

8.2 Priority II: make native orbit compression quantitative

Research problem 8.4 (A uniform trace bound for ownership-weight queries)**.** For the particular maintained weighted-orbit implementation used here, prove a polynomial upper bound on its intermediate weight runs, cycles and replay-certificate size for authenticated native ownership queries, as a function of source records and binary interval lengths.

The published weighted orbit theorem gives a polynomial algorithm for the abstract task. It does not automatically give such a bound for a different implementation's exact scheduling and proof trace. The new length field adds owner intervals that were absent from the sparse endpoint-mask input. This raises the observed run count on some cases even while removing a whole census. A correct proof must therefore track owner-interval fragmentation, not reuse a port-only sparse-token bound.

One possible route is to decompose the weight function into two channels: a small terminal channel and a structured edge-mass channel. The terminal channel has a conservation certificate; the mass channel is the image of disjoint affine owners. If native geometric constraints force a bounded number of mass breakpoints per surviving orbit phase, they may give a sharper bound than an arbitrary weighted-query analysis. If this invariant fails, a counterexample should guide a revised representation or a direct implementation of the published weighted algorithm. Maintaining exact independent replay is a useful requirement for either route.

Research problem 8.5 (Optimal deterministic pair labels)**.** Let r labels be inserted once each into an unknown partition. Find nonnegative integer labels with the smallest maximum bit length that permit exact unordered-pair decoding, a separate additive length channel, and a verifiable global inventory, without a quadratic decoding table.

The pair code in Section 5 uses a quadratic moment and a radix of order r^3. It has $O(\log r)$-bit labels and an explicit integer-square-root decoder. For a codec that must recover each pair from its scalar sum alone, the $\binom{r}{2}$ distinct unordered pairs must have distinct sums. Labels drawn from $[0, M]$ then give only $2M + 1$ possible sums, so

$$2M + 1 \geq \binom{r}{2}, \qquad M = \Omega(r^2).$$

This lower bound applies to that local additive decoding contract; it is not a lower bound for every encoding that might exploit the complete inventory globally. It leaves a factor of order r between the present magnitude and the counting bound, while the bit-length order is already optimal for the local contract. Closing this gap requires both smaller values and an efficient inverse for their pair sums.

There is a useful way to separate the inventory requirement from that inverse problem. Suppose $u_i \in [0, M]$ have distinct sums on distinct unordered pairs. Shift them to $v_i = M + 1 + u_i$. Every singleton is then at most $2M + 1$, whereas every pair sum is at least $2M + 2$. Pair-sum injectivity is unchanged. With an owner radix $D > 2 \max_i v_i$, zero- and one-label orbits cannot masquerade as a decodable pair, and the consumption argument of Lemma 5.4 applies under the same complete-histogram and multiplicity checks when the recovered matching exhausts the authenticated inventory. Thus an efficiently invertible Sidon-type construction would be a candidate; existence of distinct pair sums alone is insufficient. Any change should be benchmarked against the current power-of-two radix, whose divisions are bit shifts.

Research problem 8.6 (Compatible boundary frames and passive transport)**.** Augment the native boundary output with local fibre frames on all relevant base-boundary components, including passive components, together with the transition and orientation-obstruction data required to compare them. Prove that these data suffice for a stated gluing or surgery operation. No global fibre orientation is required when the bundle obstructs one.

The present output gives local fibre transport across active gaps and retains passive multiplicities through the optional full packet. It does not supply a chosen parametrisation of every passive boundary circle or source paths comparing all local frames. For an orientable total interval bundle, fibre holonomy around a base-boundary circle is already trivial: its vertical boundary is an annulus, not a Möbius band. Thus the missing data here are parametrisations and source-based frame comparisons, rather than an undetermined boundary fibre flip. A finite-group lift

from the incoming moment report is a candidate primitive for transport along specified source paths, including interior paths used to compare frames. It must be checked against the bundle's orientability data. Testing a few nonorientable bases is useful for finding mistakes; the main result should be a sufficiency theorem for the complete gluing record. The certificate must preserve components with no active port, because a terminal-only census cannot see them.

8.3 Priority III: bound a complete hierarchy

Research problem 8.7 (A complete quantitative hierarchy theorem)**.** Prove, or locate and verify in full detail, the precise hierarchy construction needed for a bound of $L = O((\log n)^2)$ together with the surface and pattern parameter $G = n^{O(1)}$ and polynomial local work. Then identify the additional argument required for the stronger $L = O(\log n)$ version of the announcement.

The February and March 2021 statements should be tracked separately. The 2026 preprint is a substantial source of rigorous machinery, but its iteration estimate does not furnish every bound in Theorem 7.1. A useful research artifact would map each assumption of that theorem to a proved proposition, including its actual parameter and whether it applies with empty, essential, or arbitrary boundary pattern. This would expose any missing hypothesis before implementing a search that relies on it.

Research problem 8.8 (Discover useful surfaces without exhaustive enumeration)**.** Develop a complete method to select the normal or regulated surfaces needed by that hierarchy within the target total budget. Bound the encoded normal coordinates and the number of unsuccessful candidates.

The new initial core removes certified irrelevant source handles, so it is a better input to such a search. The genus kernel may also allow the search to focus on a small number of interacting nonball pieces. Nevertheless, a single retained torus-boundary piece can already be an arbitrary knot exterior. Neither reduction permits a claim that enumerating all fundamental surfaces becomes quasipolynomial. A promising direction is a selector tied to a proved hierarchy potential, with a certificate that either supplies the required surface or justifies one of the special-region alternatives. A heuristic ranking can help experiments, but a worst-case proof must bound the fallback search too.

Research problem 8.9 (Control intersections in multisurface states)**.** Give a compressed, source-authenticated representation of the intersecting surfaces permitted by the announced multisurface method, and prove that its size remains within the state bound of Theorem 7.1.

Proposition 7.4 rules out an easy replacement by a disjoint basis. The relevant interface must encode intersection curves and arcs, their order on source pieces, and the effect of cutting in successive complements. Native first-return queries may recover some of the necessary transitions. Sparse cut grammars and disconnected tile kernels from the incoming packages may then compress repeated regions. The research burden is to prove both geometric realizability and a bound on the width or number of such regions along the actual hierarchy. A fast grammar algorithm on unrestricted input does not provide that geometric bound.

Research problem 8.10 (A combined potential for topology, interfaces and marks)**.** Find a potential that controls hierarchy depth while charging retained nonball deficits, attachment cycles, active interfaces, and the cost of the boundary pattern in one state representation.

The exact identity $\delta = \beta + \sum(1 - \chi(K_v))$ supplies an additive topological budget. The moment compiler supplies an output bound in the number of active ports and logarithmic lengths. These fit naturally into a proposed potential with separate topological and interface terms. The difficult issue is monotonicity: a valid cut can increase boundary genus and expose new ports. A successful

potential must prove that such increases are paid for by progress in the global hierarchy, or by a special-region reduction. Simply adding the available size parameters does not prove any decrease. This is a central mathematical question, not a matter of choosing convenient benchmark weights.

8.4 Priority IV: make performance evidence useful for integration

Research problem 8.11 (Choose source operators by measured structural cost)**.** Determine when the maintained boundary-period certificate, the unused component certificate, the four-field moment codec, or the compact pair codec has the best completed production-and-replay cost. Design a deterministic, inexpensive selection rule with a correctness-preserving fallback.

The delivered results already rule out an unconditional wall-clock claim. The unused-handle certificate has extra component records and can be larger on small sources; the ownership query can generate more weight runs; padded trefoil examples may become slower despite a smaller core. Candidate predictors include surviving attaching occurrences, the number of zero-degree handles, owner-run count, port count, and binary coordinate length. They should be obtained from work already required by the source compiler whenever possible. The fallback must retain exact status and certificate semantics; a timeout is not evidence of knottedness.

Research problem 8.12 (Extend independent evidence beyond the present fixtures)**.** Build a stable corpus of valid relabelled triangulations, reducible controls, high-valence source edges, large binary normal coordinates, port-free bundles, and independently known knot types. Measure both local operations and end-to-end recognition when an opt-in integration exists.

The immediate deliverable should be a source-identified record for every case, its expected theorem domain, exact proof replay, and raw samples. Separate favorable asymptotic families from finite stress tests. Retain negative cases and historical baselines. The present package follows that convention, but its finite banks are not a random sample of all knot exteriors or all normal surfaces. A broad integration evaluation should follow the mathematical and representation milestones above, so that any observed speedup can be attributed to a completed operation.

8.5 A proposed order of work

The most direct next milestone is the normal-coordinate adapter on the four initial local templates, together with an opt-in call to the authenticated initial compiler. In parallel, the ownership-weight trace bound can be pursued without changing the topology. The next geometric milestone is patterned ball absorption and compatible bundle charts. Only then is it reasonable to combine the incoming grammar and tile kernels into a recurring hierarchy state. The global depth and surface-selection arguments should be developed alongside that work, with every needed quantitative claim recorded explicitly. This ordering makes each implementation step consume a proved interface and makes failure to achieve the target exponent mathematically diagnosable.

9 Evidence, reproducibility and integration

The deliverable is organized so that a repository reviewer can assess the mathematical statement, the executable operation, and its source contract separately. The article is complete in the accompanying PDF and LaTeX sources. The new Python modules are additive research code; the pinned upstream snapshot is present to make reproduction independent of later repository changes.

9.1 Claim and implementation matrix

Result	Delivered implementation	Boundary of the claim
Genus budget and forest quotient	Producer, independent graph checker, shared source-parent forest	A graph certificate binds incidences and Euler labels. Actual ball absorption needs verified geometry and irreducibility.
Initial torus admissibility	Geometric source decomposition, independent raw replay, exact canonical-diagram wrapper	Empty initial pattern and inherited handle charts. The intermediate core is not a knotted verdict.
Occurrence replay and Fibonacci separation	Counter-based replay, symbolic-family audit and explicit certificate comparisons	Linear incidence updates exclude source reconstruction; the size separation compares the maintained explicit format.
Native first returns	Pair and four-field codecs, independent source checker, complete cyclic assembly, exact gap lengths	Local fibre data and active words; arbitrary later markings and complete base-frame transport remain outside the contract.
Global inventory lemma	Exhaustive eight-label partition regression and the native decoder's coverage checks	A complete authenticated weighted histogram is assumed. Arithmetic pairs alone do not establish geometry.
Unmarked reconstruction	Written constructive theorem and explicit output-size proof	No reconstruction implementation or fixed marking transport is delivered.
Forest transport circuit	Written algebraic construction	Source parent paths are implemented; arbitrary group labels and geometric chart transport are not.
Quasipolynomial accounting	Conditional theorem and research questions	The required complete hierarchy, discovery bounds and pattern-size bounds are not established by this package.

9.2 Final combined verification

The final combined run passed all **30 test methods**, with no failures, errors or skips, in 113.179 seconds. Regina 7.4 was available, so its independent topology controls ran. The exact log is `results/validation_log.txt`; the machine-readable environment and result are in `results/validation_summary.json`.

The graph tests enumerate all 772 connected labelled simple graphs with one through five vertices and test three root assignments on each, for 2316 graph/root cases. They also check 500 seeded random multigraphs, sharp genus examples, parent routes, loops, parallel edges, disconnected input rejection, altered forest and root records, and strict integer typing. Boolean values and floats equal to integers are not accepted as integer proof fields. The additional source-to-kernel test authenticates all 48 supplied decompositions, passes them through the adapter, and checks their graph certificates independently. This test explicitly preserves the generic, non-recognition status on reducible controls.

The source-handle tests compare the new deletion trace and remaining handle sets against the maintained reducer on all 48 raw source cases. They also perform independent tetrahedron and local-vertex relabellings, exercise the canonical unknot, trefoil and figure-eight diagram sources, and

reject forged source provenance. The reducible manifolds $(S^1 \times D^2)\#\mathbb{RP}^3$ and $(S^1 \times D^2)\#(S^2 \times S^1)$ are central controls: their numerical cycle signatures pass the generic decomposition checker, but are not solid tori. Independent Regina tests and the diagram-source gate keep the distinction executable.

The native suite compares both new codecs with the complete maintained cyclic interface on 157 recorded triangulation/normal-vector pairs. An independent literal edge walk covers 146 of them. Its 219 positive gaps and 3420 point gaps test exact length, while the cap-only cases and whole-circle attachments exercise source geometry that a prism-only adapter would miss. Resource caps return an inconclusive operation; they never turn incomplete work into a partial accepted certificate. The tests also disable discovery routines during replay, alter source fingerprints and proof fields, and test JSON round trips for large integers. The separate 4140-partition enumeration checks the arithmetic occupancy lemma without relying on a geometric path promise.

These are finite implementation checks. The asymptotic claims rest on the proofs in the article. The package does not claim formal verification of Python, statistical representativeness of the fixtures, or complete testing of unrelated maintained recognition subsystems.

9.3 What the measurements establish

All source and native tables retain the individual timing samples in machine-readable JSON. There are five measurements per arm and source; the native comparison has an untimed warmup and alternating forward and reverse arm order, while the source-handle comparison uses seeded randomized order. The graph-only experiment uses seven repetitions after one warmup. Measurements were made in a shared execution environment, with Python 3.12.14 on Linux x86-64. They are medians of finite samples, without a claim of hardware isolation or a statistical confidence interval for a population of knot diagrams.

The new native compiler sometimes improves full discovery and replay together, and sometimes improves discovery while making replay slower. Its stronger invariants are exact: one census instead of two, logarithmic terminal labels, exact additional lengths, and a smaller proved scalar magnitude for the pair codec. The new source-handle certificate has an asymptotic size advantage on its proved family, but extra component-record overhead can make it larger at small sizes. The canonical trefoil and the 256-bud trefoil show measured source-operator regressions. These results support an opt-in integration and further cost analysis; they do not support a universal speedup claim.

The graph experiment isolates the cost of retaining source provenance while shrinking the operational graph. Its realizable examples contain three solid-torus-type vertices and an increasingly long chain of ball-type vertices. The kernel always has three vertices and two edges; the certificate retains the full source-parent forest.

Source vertices	Kernel vertices/edges	Production, ms	Replay, ms	Proof bytes
10	3/2	0.034	0.044	593
100	3/2	0.252	0.335	2531
1000	3/2	3.003	3.499	26879
10000	3/2	36.015	44.774	319877

Table 7: Graph-only genus-three kernels with full source provenance, seven repetitions. Geometric source authentication and disc-chart construction are excluded.

This experiment excludes construction or authentication of the vertex manifolds and their disc charts. It measures the graph routine described in its theorem, not a hidden geometric ball-recognition algorithm.

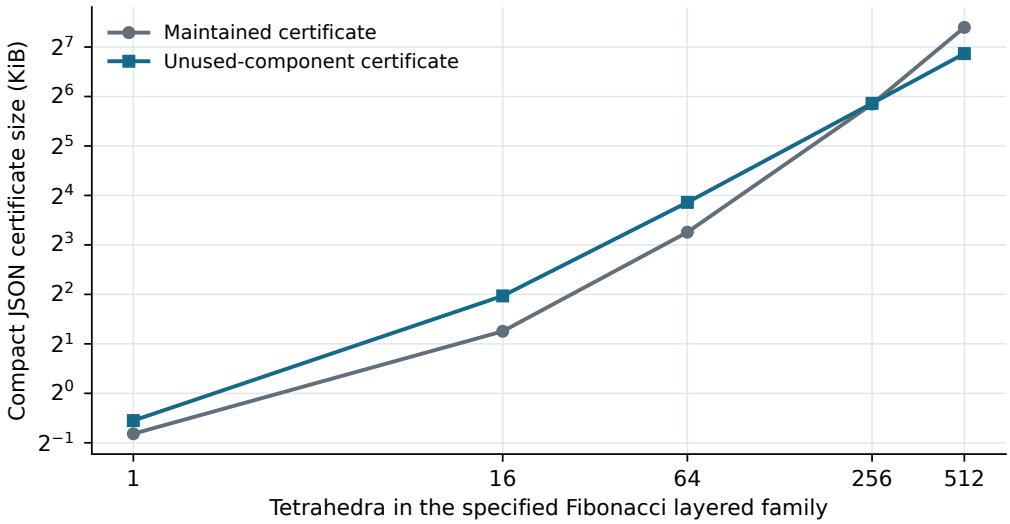


Figure 2: Recorded compact-JSON certificate sizes on the specified Fibonacci layered family. Lines connect measured points only. Small instances expose the new format's overhead; the proof of its asymptotic advantage is Proposition 4.6, not a fit to this plot.

9.4 Package layout and replay

The principal executable interfaces are as follows.

`decompose_unused_handles`
: Consumes a validated raw torus-boundary source. Returns a geometric decomposition and a numerical case, with no irreducibility assertion. `verify_unused_handle_certificate` reconstructs source cells and replays the trace independently.

`compile_diagram_initial_core`
: Consumes a classical knot diagram and derives its canonical exterior. Its checked outputs are `UNKNOT`, `ADMISSIBLE_CORE`, or resource-limited `INCONCLUSIVE`. The independent wrapper checks the exact diagram and exterior provenance before using the genus theorem.

`kernel_from_decomposition`
: Converts the source decomposition format into a rooted-forest graph certificate. It is an adapter, not a substitute for the geometric checker. The independent graph checker accepts the original Euler labels and edge incidences alongside the proposed certificate.

`normal_parallelity_moment_words`
: Consumes a triangulation and binary normal coordinates, and returns the complete native cyclic interface with exact lengths. The independent checker rebuilds the source through the reference path. The optional `moments_from_parallelity_packet` entry point additionally verifies and retains a complete upstream parallelity certificate.

From the archive root, the following commands run the complete delivered suite and small verified examples:

```
python3 -m pip install -r requirements.txt
python3 run_checks.py
python3 examples.py
```

The test runner inserts the new code and pinned native snapshot into its import path. The examples show a positive unknot certificate, the intermediate trefoil core, and an exact native boundary query. Representative native certificates can also be replayed without running discovery, using `code/native_moments_replay.py` with the import path documented in the root README.

The new algorithms use Python's standard library. Regina 7.4 is listed to reproduce the separate topology controls in the complete recorded run. Matplotlib is needed only to regenerate the figures, which are already included. A TeX distribution builds the article using `latexmk -pdf hierarchy_primitives.tex` from its article directory. No private external data source is needed for reproduction.

9.5 Suggested repository integration

The initial integration should preserve the package as an additive research contribution, with its exact source revision and evidence. `integration/INTEGRATION.md` maps the modules to a proposed maintained placement and records the private source APIs they currently use. The pinned source snapshot is for reproduction; an integrated tree should use the repository's matching maintained modules instead of maintaining a second copy of them.

For a first opt-in recognizer change, call the canonical-diagram initial compiler at the same stage as the existing source-handle operator. Accept a positive result only through the diagram-bound checker. A returned core should enter the next stage only once that stage supports the inherited handle presentation and its normal-coordinate interface. For the boundary compiler, compare full old and new outputs in a differential integration mode, then select the new operator only under its documented source contract. Preserve caps and interruption behavior.

No default dispatch or remote repository mutation is included in this deliverable. The package supplies concrete code and proofs for review; the unresolved patterned and global interfaces remain explicit research tasks. The goal of the proposed integration is to reduce real local work while keeping every eventual recognition claim attached to its geometric authority and complete complexity accounting.

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